The International Journal of Life Cycle Assessment (2025) 30:405–428
https://doi.org/10.1007/s11367-024-02409-4

CRITICAL REVIEW

Digital technologies for life cycle assessment: a review and integrated combination framework

Martin Popowicz[1] · Nicolas J. Katzer[1] · Moritz Kettele[1] · Josef-Peter Schöggl[1,2] · Rupert J. Baumgartner[1]

Received: 2 July 2024 / Accepted: 12 November 2024 / Published online: 7 December 2024

Abstract

Purpose Companies need to enhance their understanding of the environmental impacts of their products and services. Life cycle assessment (LCA) has become a prevalent method for evaluating these impacts. Despite significant advancements in LCA methodology and data availability, several challenges persist. Digital technologies may offer solutions to these challenges in LCA. Therefore, it is crucial to explore how digital technologies can be integrated into LCAs.

Methods A systematic literature review was conducted to examine the application of digital technologies, specifically blockchain, the Internet of Things (IoT), big data, and artificial intelligence (AI), within LCAs. The review included 103 peer-reviewed journal articles and conference papers. Contributions of these technologies were categorized according to the four LCA phases outlined in ISO 14040/44 standards. The findings were synthesized into a framework that highlights the individual and combined potential of these technologies for enhancing LCAs.

Results and discussion The review reveals that IoT is primarily used in the inventory analysis phase, while blockchain, AI, and big data are applied across the goal and scope definition, inventory analysis, impact assessment, and interpretation phases. Based on these findings, a comprehensive theoretical concept was developed to outline all possible combinations of these four technologies with LCA for synergistic application.

Conclusions This study proposes a framework for integrating four key digital technologies—blockchain, IoT, big data, and AI—into LCAs to support environmental sustainability assessment from a company perspective. This framework offers a current overview and a foundation for future research. For LCA practitioners, it serves as a strategic tool for identifying potential technologies and making informed decisions about which digital technologies to apply in their assessments.

Keywords LCA · Life cycle assessment · Artificial intelligence · Blockchain · Internet of Things · Big data

1 Introduction

The increasingly tangible effects of climate change and growing public awareness have heightened the need for companies to improve the sustainability of their products and services and to effectively communicate their sustainability performance to stakeholders. Life cycle assessment (LCA) has emerged as a widely used tool for evaluating the environmental impacts of products and services. LCA addresses various environmental issues and serves as a valuable decision-making tool for companies (Jolliet et al. 2015). Its strengths in this area have contributed to its growing popularity (Moutik et al. 2023). Despite its widespread use and ongoing improvements in methodology, LCA still faces several challenges. One notable challenge is data collection for the life cycle inventory (LCI) (Testa et al. 2016), which is estimated to account for 70% to 80% of the time and costs involved in conducting an LCA (Miah et al. 2018). Additionally, inaccuracies and gaps in data can introduce uncertainty into LCA models (Hauschild et al. 2018). Another challenge is the availability of data for life cycle impact assessment (LCIA) (Curran 2014), which requires linking consumption and emissions to impact categories. Some impact models are

Communicated by Matthias Finkbeiner.

✉ Martin Popowicz
martin.popowicz@uni-graz.at

1 Christian Doppler Laboratory for Sustainable Product Management Enabling a Circular Economy, Department of Environmental Systems Sciences, University of Graz, Merangasse 18/I, 8010 Graz, Austria

2 School of Sustainability, Governance, and Methods, Modul University Vienna, Am Kahlenberg 1, 1190 Vienna, Austria

not fully developed, and certain impact data remains unavailable (Chandrakumar and McLaren 2018). For example, there are still data gaps in toxicity-related impact categories, highlighting ongoing modeling limitations regarding human and ecotoxicity (Fantke et al. 2021). Furthermore, validating the information used in LCA is crucial for ensuring credibility, especially when dealing with external data that may have unknown technological or spatiotemporal relevance (Teh et al. 2020; Hauschild et al. 2018; Zargar et al. 2022). Given these challenges that affect the execution and reliability of LCAs, digitalization offers potential solutions. This study focuses on identifying how digital technologies can facilitate the execution of LCAs. Specifically, it examines the following four technologies: the Internet of Things (IoT), big data, artificial intelligence (AI), and blockchain. This selection was inspired by the technologies the European Commission (2020) identified as crucial for advancing a more circular economy. Given their importance in this context, it was considered appropriate to investigate their application in environmental sustainability assessments, particularly in LCA. Therefore, the research questions addressed are:

RQ 1: How are IoT, big data, AI, and blockchain applied within LCAs?
RQ 2: How can these technologies, from an organizational perspective, be combined to assess environmental performance?

To answer these questions, a systematic literature review was conducted. The results, including the various applications of the technologies, are categorized according to the four phases of LCA as defined by ISO 14040/44 standards. To the authors' knowledge, previous studies have generally focused on one specific digital technology within the LCA context (Ghoroghi et al. 2022; Karaszewski et al. 2021). In contrast, this review not only includes an examination of all four technologies but also explores their individual and combined applications. Additionally, it proposes a framework for integrating all four technologies to enhance LCA practices.

The remainder of the paper is organized as follows. Section 2 provides the theoretical background, Sect. 3 details the methodology used to identify and analyze relevant literature, and Sect. 4 presents the results of the systematic literature review. The paper concludes with Sects. 5 and 6, which cover the discussion and conclusions, respectively.

2 Theoretical background

2.1 Life cycle assessment and its challenges

LCA is governed by ISO 14040/44 standards, which define it as the "*compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product system throughout its life cycle*" (ISO 2006a, p. 7; ISO 2006b, p. 8). The standards outline four distinct phases of conducting an LCA: *goal and scope* definition, *inventory analysis*, *impact assessment*, and *interpretation*. Traditionally, LCAs quantify potential environmental impacts such as global warming, eutrophication, and terrestrial ecotoxicity, encompassing a broad range of environmental issues (Jolliet et al. 2015). Complementary assessments, such as social life cycle assessment (S-LCA) and life cycle costing (LCC), extend the focus to other sustainability dimensions, contributing to a comprehensive life cycle sustainability assessment (LCSA). However, no standardized framework for LCSA currently exists (Larsen et al. 2022).

LCA's broad application spectrum and its effectiveness as a decision-support tool have contributed to its growing popularity (Moutik et al. 2023). Despite this, several challenges remain, including applicability for practitioners, communication of results to the public, methodological improvements for researchers, and the structuring of new data sources for data providers (Mutel 2023). While digital technologies offer solutions to these challenges, only the first challenge is of major interest from the company perspective taken in this publication. This challenge mainly relates to the data collection for the LCI (Testa et al. 2016).

According to Hauschild et al. (2018), collecting all relevant LCI data for the reference system can be organized into two main categories: the foreground system and the background system. The foreground system consists of *physically embodied* processes, such as the directly utilized materials that contribute to the reference flow, and *support function* processes, such as the energy inputs required to operate the system. This system is directly influenced by and should be well understood by the focal company. In contrast, the background system includes *deliver service* processes, such as auxiliary processes and market trends that indirectly impact the system, as well as *infrastructure* processes, such as the background system of the subject under investigation. These are not under the direct control of the focal company. Thus, while robust primary data is crucial for both the foreground and background systems, the foreground system is potentially traceable using only in-house resources and competencies, which can provide the most accurate and specific LCI data (Hauschild et al. 2018). However, tracking and documenting resources, emissions, and energy streams that are not yet automatically and digitally recorded can be challenging and, in some cases, may prove to be inaccessible (Saavedra-Rubio et al. 2022).

Therefore, data that is not directly available often requires external assistance from experts and related publications, such as industry representatives, consultants, environmental reports, legal documents, statistical agencies, consumer organizations, or scientific publications (Hauschild et al.

2018). Additionally, specialized LCI databases, like Ecoinvent (Ecoinvent 2024) and GaBi (Sphera 2024), are commonly used by professionals to obtain data not available in-house (Kiemel et al. 2022). These databases primarily rely on manually managed primary datasets to ensure data quality and facilitate process matching through tailored search engines (Zargar et al. 2022). However, this reliance on manually curated data presents a limitation: the databases may not fully capture the technological and spatiotemporal specifics of every anthropogenic process (Mutel 2023). Finding well-suited datasets can be challenging because there is often no perfect match between the highly specific process of the system under investigation and the available LCI data. Data may be outdated, not reflecting current technological advancements, or generalized, covering different regions, models, or scenarios than those being evaluated. Methods like the Pedigree Matrix (Ciroth et al. 2016) help assess the quality of such datasets, identifying those that are not suitable for the specific system under study. However, such unfitting data may introduce another time-consuming applicational step.

If the required data is unavailable in the needed quality or with specific process details, or if there are data gaps where the data is entirely missing, the focal company applying the LCA must develop or find a suitable model to estimate the input and output inventory of the affected process(es). To address these gaps, process models can be constructed using data proxies—by modifying surrogate (sub-)processes to represent the missing data—or data creation methods, such as empirical equations or mechanistic models based on natural sciences (Zargar et al. 2022). These approaches help account for technological developments, date, and geographic differences to supplement the missing data in the system model (Zargar et al. 2022).

In conclusion, it is best practice to accompany the inventory analysis phase of an LCA with a sensitivity analysis. This helps evaluate any remaining uncertainties related to the model, parameters, and scenarios used in the assessment (Bamber et al. 2020). It is estimated that approximately 70% to 80% of the time and costs associated with conducting an LCA are attributed to data collection and linking various value chains for the LCI (Miah et al. 2018). However, the advancement of digital technologies can be a powerful tool to address these challenges not only in the inventory analysis phase but also in other phases of an LCA, including those discussed in the following sections.

2.2 Digital technologies

Digitalization, which refers to the integration of information and communication technologies into everyday life, society, and the economy (Gossen et al. 2021), brings significant changes and presents both challenges and opportunities for companies (Gimpel and Röglinger 2015). Digital technologies, as key drivers of digitalization, play a crucial role in the digital era (Berger et al. 2018) and can enhance the circularity of products. The European Commission (2020) highlights four technologies, IoT, big data, AI, and blockchain, as particularly influential in this regard. This study focuses on these technologies and their benefits for LCA.

- IoT is defined as: "A group of infrastructures interconnecting connected objects and allowing their management, data mining, and access to the data they generate" (Dorsemaine et al. 2015, p. 73).
- Big data is characterized by three Vs: Volume (the extensive amount of data), variety (the diversity of data, including both structured and unstructured data), and velocity (the speed at which data is generated) (Fasel and Meier 2016).
- AI, as defined by Nilsson (2010, p. xiii), "is that activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment."
- Finally, blockchain is described as: "a distributed database of records or public ledger of all transactions or digital events that have been executed and shared among participating parties. Each transaction in the public ledger is verified by consensus of a majority of the participants in the system. And, once entered, information can never be erased. The blockchain contains a certain and verifiable record of every single transaction ever made" (Crosby et al. 2016, p.8).

3 Materials and methods

To address the research questions, a systematic literature review was conducted following the "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA) guidelines (Moher et al. 2009). The review focuses on the application of IoT, blockchain, AI, and big data in LCA, where this section provides an overview of the process used to identify and include literature in the review. Further, the qualitative content analysis and the development of the integrated framework are described.

Scopus and Web of Science (WoS) were selected as search databases, and the search terms used are listed in Table 1. The initial search terms were chosen after thorough testing of various options, as they led to the most accurate results. While the use of common acronyms (e.g., LCA or AI) might have led to a higher number of search results, their inclusion during the trials led to less accurate results, given the ambiguity of these acronyms. An exception was the acronym IoT, and hence, it was included. To address this potential shortcoming of excluding the acronyms, the search

Table 1 Search terms used

Technology	Search term
Internet of things	TITLE-ABS-KEY ("life cycle assessment" AND ("internet of things" OR "IoT"))
Blockchain	TITLE-ABS-KEY ("life cycle assessment" AND "blockchain")
Artificial intelligence	TITLE-ABS-KEY ("life cycle assessment" AND ("artificial intelligence" OR "machine learning"))
Big data	TITLE-ABS-KEY ("life cycle assessment" AND "big data")

not only incorporated keywords but used titles, abstracts, and keywords to ensure that relevant studies were identified. Using the initial search terms, 481 potentially relevant papers were identified: 122 from WoS and 359 from Scopus. The initial search was conducted in March 2023, with no restrictions applied, to ensure that no potential applications were excluded based on geographical context or time horizon. After removing duplicates, 329 papers remained for screening. These 329 papers underwent a title and abstract screening, during which papers not related to the application of the mentioned technologies in LCAs were excluded. This process resulted in the exclusion of 145 papers, leaving 184 papers for full-text screening. During full-text screening, the same exclusion criteria were applied, but only journal and conference papers were considered. This focus on journal articles and conference papers was chosen to, on the one hand, include only peer-reviewed publications and, on the other hand, also include the latest developments that have so far only been published as conference proceedings. Therefore, the approach of focusing on conference papers and journal articles was seen as suitable, and hence, books and book chapters were excluded from the analysis. Further, also literature reviews were excluded from the sample but were used for extensive snowballing instead. After full-text screening, 74 papers remained. Additionally, 29 papers identified through snowballing were deemed suitable based on the same criteria, bringing the final sample to 103 papers for qualitative content analysis. Summarizing, the inclusion criteria for the literature screening process were the application of one of the relevant technologies within the ISO framework for LCA (ISO standards 14,040/44). Figure 1 summarizes the literature screening process.

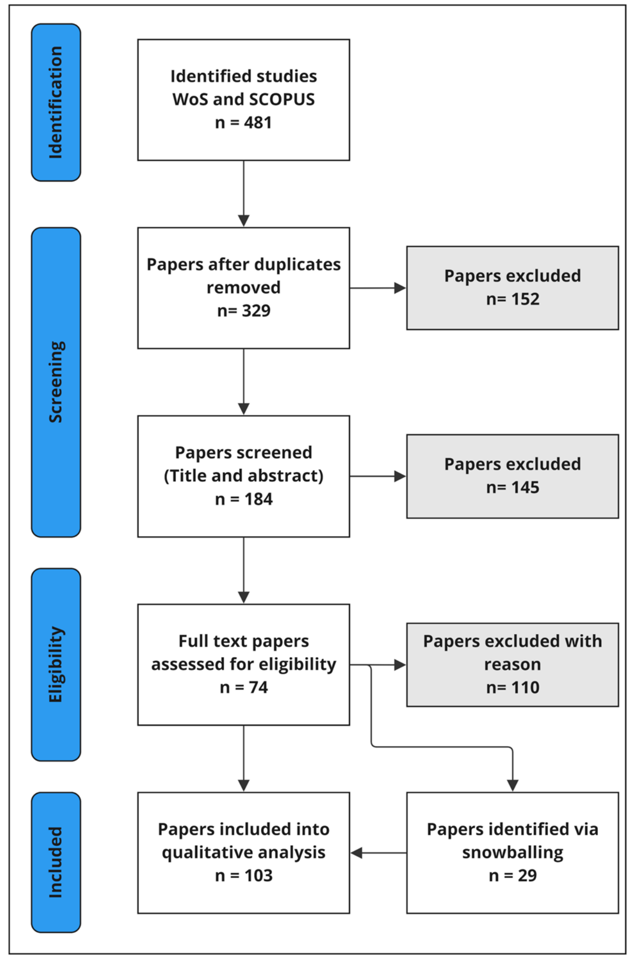

Fig. 1 Literature review process

The in-depth qualitative content analysis followed the approach outlined by Mayring (2014), utilizing the MAXQDA software package. An inductive category formation approach was chosen as it was deemed suitable for the study's goal: to identify and categorize various applications of digital technologies within LCA and summarize them into distinct categories. The category formation was based on how each technology could assist in performing an LCA and its application. The identified applications were grouped into inductively formulated categories, which were continuously updated and refined during the coding process. Based on the interpretation of results from this inductive category formation process, the final applications of digital technologies for LCA were determined. These distinctive application types were then assigned to different phases of an LCA according to ISO 14040/44 standards. The results of this first step are shown in Figs. 2, 3, 4, and 5.

In the second step, a conceptual framework was developed based on potential combinations proposed in the identified literature. To illustrate applications focusing on data exchange along the supply chain, a supply chain perspective was adopted, with the producer of the final product identified as the LCA-conducting company. The producer was chosen as

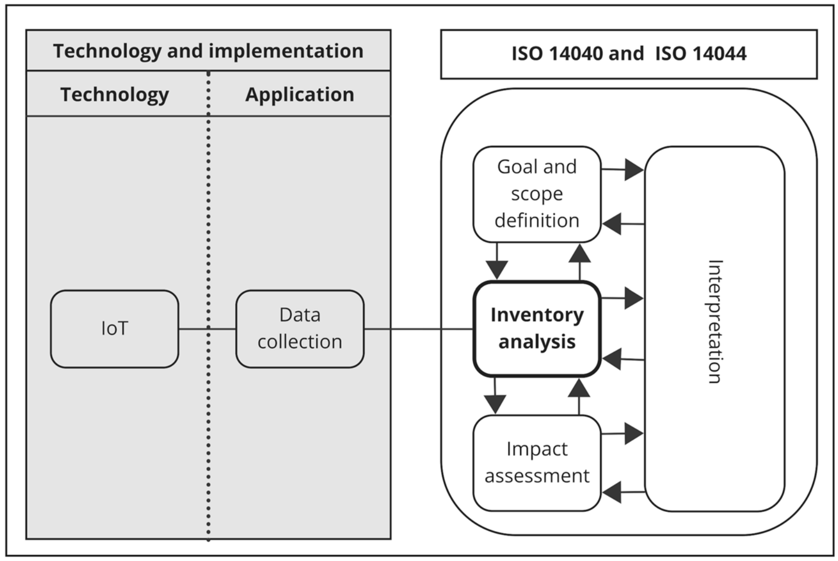

Fig. 2 Life cycle assessment and Internet of Things

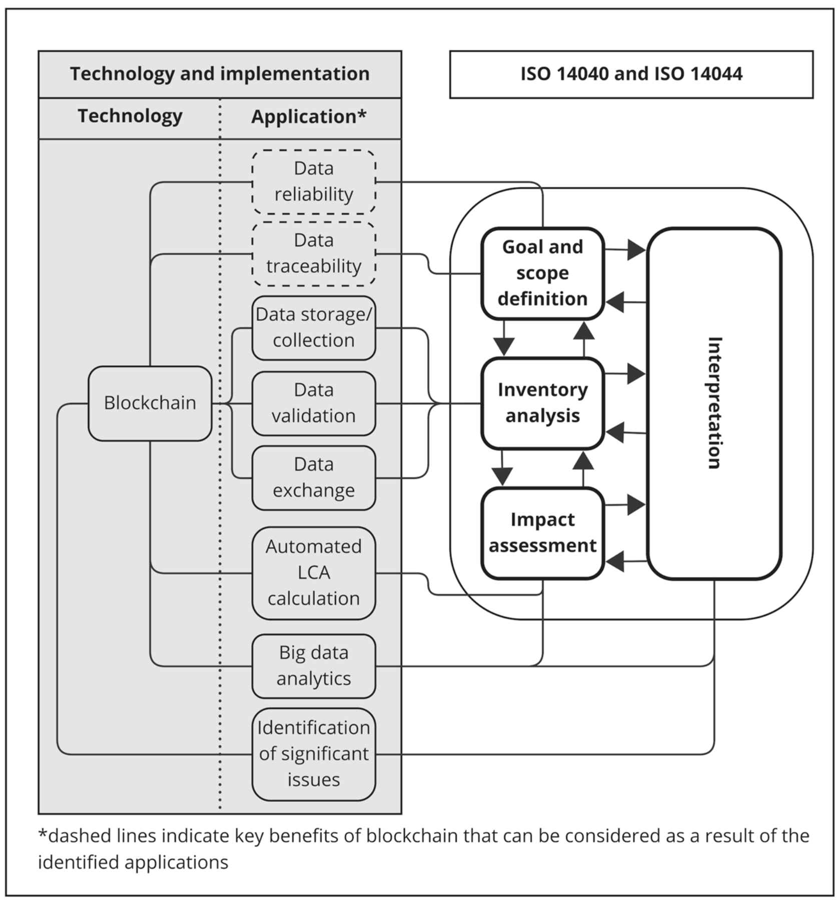

Fig. 3 Life cycle assessment and blockchain

an example due to its critical role in the supply chain. As a key actor, the producer collects and consolidates information from raw material and intermediate product suppliers and has access to data on the distribution of the end product, and potentially consumer information. Additionally, the developed framework was validated by the authors' research group through ongoing feedback throughout the development process. The framework resulting from this second step is shown in Fig. 6.

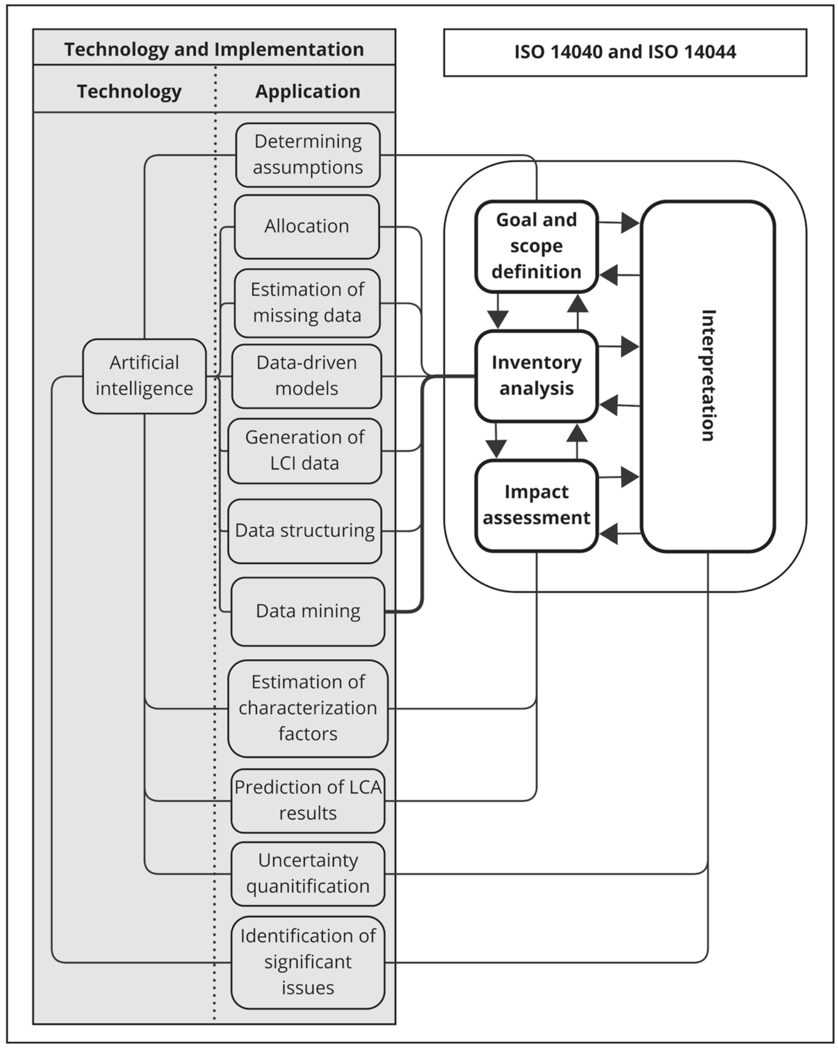

Fig. 4 Life cycle assessment and artificial intelligence

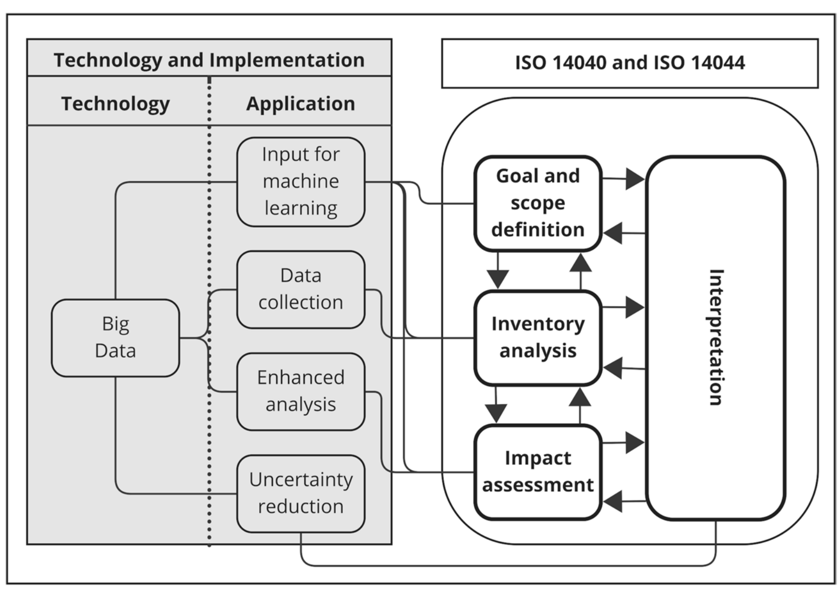

Fig. 5 Life cycle assessment and big data

4 Results

In this section, the potential applications of the investigated digital technologies are first presented individually, followed by the introduction of a framework for combining the four technologies. Papers may include the use of more than one technology and, as a result, may appear in multiple subchapters or results tables.

4.1 Life cycle assessment and Internet of Things

Out of the 103 papers reviewed, 17 focused on the application of IoT in LCA. IoT technologies are used exclusively for data collection and are therefore applied during the inventory analysis phase according to ISO 14040/44 standards. For LCA the significance of IoT, with its ability to enable data mining and access to data generated by interconnected objects (Dorsemaine et al. 2015), lies particularly in its potential to collect primary data, thus facilitating the use of accurate, real-world data for conducting more precise LCA. Figure 2 summarizes the applications of IoT for LCA, identified in the qualitative content analysis described in Sect. 3.

Three studies address the use of IoT for LCA calculation in the context of manufacturing (Mashhadi and Behdad 2018; Mishra and Singh 2019; Garcia-Muiña et al. 2018). Mashhadi and Behdad (2018) study the potential environmental impacts of Industry 4.0 and smart manufacturing, as well as the emerging possibilities for LCA. They suggest that the concept of self-awareness, originally related to monitoring, assessing, and predicting the health status of machines, can be enhanced by considering environmental impacts. Based on the data collection and interconnectedness provided by IoT, they propose the concept of "ubiquitous LCA," enabling the relaxation of assumptions regarding the linearization of heterogeneous attributes (e.g., use-phase consumer energy consumption or variable power demands) as impact calculations can then be performed using specific identity data for any single product or process. Mishra and Singh (2019) develop a framework for carbon emission reduction and sequestration within the manufacturing industry and use LCA to track carbon emissions throughout the product development life cycle. They use IoT technologies to collect real-time emission data generated during the product's life cycle. For inventory data collection, they introduce a smart manufacturing network covering physical devices in the design phase, machines in the manufacturing phase, and transportation. In addition to its use for evaluating carbon emissions based on LCA and identifying emission reduction potential in the manufacturing sector (Mishra and Singh 2019), IoT's ability to reduce carbon emissions is also recognized in the context of manufacturing of prefabricated components for construction (Tao et al. 2018) and also in other areas, such as smart cities (Asopa et al. 2021), and food industry (Luo et al. 2022). Garcia-Muiña et al. (2018) investigate how manufacturing companies can integrate sustainability principles into their business and state that IoT technologies can be used to collect process data for building databases for LCAs. In their case study, they store the data collected via sensors in the company's enterprise resource planning system, which then served as a suitable database for LCAs.

Three studies focus on the application of IoT devices for data collection in agriculture, food production, and retail (Ma and Kim 2015; Shakhbulatov et al. 2019; Asif and Gill 2022). Ma and Kim (2015) address use phase modeling in LCA and propose a usage modeling technique that utilizes sensor data as input, enabling real-time predictive LCA and leading to more accurate results. They also conduct a case study in the agricultural sector. Shakhbulatov et al. (2019) concentrate on tracking the carbon footprint of food production and propose using IoT devices to collect data relevant to carbon footprint calculations directly from trucks. Asif and Gill (2022) focus on developing a framework for blockchain in green supply chain management, using the supply chain of supermarket food products as an example, and also recognize the role of IoT in automatic data collection during the inventory analysis phase.

In the energy sector, Adedeji et al. (2020) examine the use of AI in LCI for renewable energy systems and acknowledge the contribution of sensor technologies to acquiring primary data. An et al. (2021) design an LCA platform for wind turbines based on IoT architecture, utilizing IoT technology for real-time collection of energy consumption and environmental impact data throughout the life cycle of wind turbines.

Two studies explore IoT for LCA in fashion and textiles (Shou and Domenech 2022; Chit et al. 2021). Shou and Domenech (2022) focus on using blockchain and LCA to enable circular fashion strategies for leather handbags and highlight the potential of IoT for primary data collection at the facility level. Chit et al. (2021) propose a framework for IoT and data-driven sustainability assessment in the textiles industry, developing a four-layered solution for integrating IoT technologies, particularly at the product manufacturing stage.

The remaining studies are more diverse. Shojaei et al. (2020) discuss blockchain for enhancing built asset sustainability and mention data collection via sensors and IoT as a means of improving transparency. Ingrao et al. (2021) focus on chemical formulation production and suggest that IoT technologies offer more specific data for conducting LCAs.

They use sensors to measure machine energy consumption directly for attributional LCA and compare these results with two more generic models. They find that more generic models tend to overestimate energy consumption (e.g., 74% higher compared to specific data), leading to higher LCA results. Ross and Cheah (2019) explore the benefits of high-granularity use-stage data for LCA, employing sensors to collect this data. Rolinck et al. (2021) develop a blockchain-based data management concept to facilitate LCA in aircraft maintenance, repair, and overhaul, concluding that their prototype enables automated data collection via IoT from manufacturing and maintenance processes. Tao et al. (2014) suggest an LCA calculation based on the bill of materials (BOM) and IoT technologies, focusing on energy-saving and emission-reduction (ESER) LCA. They introduce the "big BOM" data storage concept for integrating their developed ESER evaluation system with data from various company information systems. The proposed concept serves as a data source for the ESER evaluation tool and combines two types of data: (1) available BOM information within the enterprise and (2) energy consumption and environmental impact data generated throughout the entire product life cycle. The BOM data is extracted from existing enterprise information systems (e.g., product data management, enterprise resource planning, computer-aided design/process planning, supply chain management) and IoT technology is used for collecting energy consumption and environmental impact data across the product life cycle. Finally, Zhang et al. (2020) mainly focus on a framework for blockchain-based LCA but also propose integrating IoT for automatic and real-time data collection during the inventory analysis phase. Often, IoT is used and managed centrally, whereas Van Capelleveen et al. (2018) develop a decentralized concept for storing LCA-related data collected via IoT. Their system, called the "footprint of things," proposes storing data directly on product components to enable instance-specific inventory data collection rather than using average values for entire product lines. Essentially, it is a hybrid system consisting of decentralized sensors and storage nodes paired with a centralized data repository, where average LCI data can be stored. Based on the LCI data recorded on the components, an LCA of a product can then be calculated using the stored data.

Table S1 in the supplementary information summarizes the IoT-focused papers from the literature sample.

4.2 Life cycle assessment and blockchain

Overall, 14 out of the 103 included papers addressed the use of blockchain in the LCA process. When implementing blockchain, organizations need to decide whether to implement a centralized or decentralized database solution, both of which have various advantages and disadvantages (Islam and Apu 2024). Decentralized databases, which are commonly used in public blockchains, such as cryptocurrencies, enhance security and transparency by removing the need for a central authority (Islam and Apu 2024). Thus, they reduce the risk of data manipulation, fostering trust among participants, but face limitations in scalability and efficiency due to slower transaction speeds and higher resource demands (Islam and Apu 2024). Centralized databases, which have a central authority and are typically found in private or permissioned blockchains, provide enhanced efficiency and scalability and faster transactions, but also introduce vulnerabilities such as single points of failure and decreased transparency, which can undermine trust and data integrity (Islam and Apu 2024). These trade-offs must be considered when implementing blockchain for LCA, which is utilized across all four phases according to ISO 14040/44 standards. In the goal and scope definition phase, blockchain impacts the definition of data requirements. During the inventory analysis phase, blockchain is employed for data storage, collection, and exchange. This leads to enhanced reliability and traceability of data, as validations are enabled for all nodes (i.e., stakeholders) of the blockchain. In the impact assessment phase, blockchain can facilitate big data analytics and automated LCA calculations, while in the interpretation phase, it can assist in identifying key issues. Figure 3 summarizes the applications of blockchain for LCA identified in the qualitative content analysis (Sect. 3). Since enhanced reliability and traceability can rather be considered benefits based on the application of blockchain, they are indicated with dashed lines in Fig. 3.

Blockchain's primary contribution to LCA is in the goal and scope definition phase, where it enhances data reliability and traceability. This potential benefit is noted by all 14 publications (Zhang et al. 2020; Teh et al. 2020; Shojaei et al. 2020; Shakhbulatov et al. 2019; Rolinck et al. 2021; Asif and Gill 2022; Shou and Domenech 2022; Lin et al. 2021; Li and Wang 2021; Carrières et al. 2022; Chit et al. 2021; Van Capelleveen et al. 2018; Mishra and Singh 2019; An et al. 2021). Studies dealing with the IoT also recognize this advantage. Chit et al. (2021) and Mishra and Singh (2019) suggest the integration of blockchain into their IoT-enabled frameworks for LCA in order to enhance data reliability. An et al. (2021) and Van Capelleveen et al. (2018) also acknowledge the potential future integration of their IoT-focused approaches with blockchain.

In the inventory analysis phase, the two most relevant applications of blockchain are data storage and collection (Zhang et al. 2020; Shojaei et al. 2020; Shakhbulatov et al. 2019; Rolinck et al. 2021; Asif and Gill 2022; Shou and Domenech 2022; Lin et al. 2021; Li and Wang 2021; Carrières et al. 2022; Chit et al. 2021; Mishra and Singh 2019), as well as the closely related application of data exchange (Zhang et al. 2020; Shojaei et al. 2020; Shakhbulatov et al. 2019; Rolinck et al. 2021; Asif and Gill 2022; Shou and

Domenech 2022; Lin et al. 2021; Li and Wang 2021; Chit et al. 2021).

Three studies dealing with data storage and collection and data exchange in the inventory analysis phase focus on the textiles and fashion industry (Chit et al. 2021; Shou and Domenech 2022; Carrières et al. 2022). Shou and Domenech (2022) propose a framework for integrating blockchain into LCA, using leather handbags as a case study. Carrières et al. (2022) explore how a blockchain traceability platform can improve LCA results for textile products. They first calculate a generic LCA and compare the results to a specific LCA based on data from the blockchain traceability platform. The comparison reveals significant differences between the specific and generic LCAs. Finally, Chit et al. (2021) refer to blockchain as a possibility for the storage of data collected via IoT.

Two studies focus on the application of blockchain for LCA in the built environment (Shojaei et al. 2020; Li and Wang 2021). Shojaei et al. (2020) examine the use of blockchain to track data necessary for sustainability assessments in the building industry. Their proposed information chain starts with the extraction of raw materials and extends to the final product reaching the vendors. Li and Wang (2021) focus on blockchain and the circular economy in the architecture, engineering, and construction industry, emphasizing the role of blockchain in data collection and exchange for LCA.

In the context of manufacturing, Lin et al. (2021) propose a framework for blockchain-based LCA utilizing blockchain technology for the secure transmission of inventory data from upstream suppliers to downstream manufacturers. Mishra and Singh (2019) include blockchain as possibility for data storage in their IoT-enabled LCA framework.

The remaining studies, focusing on the applications data storage and collection and data exchange, are more diverse. Shakhbulatov et al. (2019) investigate the use of blockchain to track the carbon footprint within food supply chains, particularly focusing on the transportation stage. They propose a system called the Carbon Footprint Chain, a lightweight, scalable blockchain implementation. Asif and Gill (2022) propose that blockchain can facilitate LCA execution, particularly for LCI, by enabling real-time data collection via IoT and supporting LCIA through the combination of IoT and blockchain for handling large data volumes. Rolinck et al. (2021) create a blockchain-based data management concept to facilitate LCA, applying it in the context of aircraft maintenance, repair, and overhaul. Finally, Zhang et al. (2020) develop a framework for blockchain-based LCA, integrating blockchain with IoT and big data.

The third identified application in the inventory analysis phase is data validation. For instance, Shou and Domenech (2022) propose various data validation technologies within the blockchain-LCA framework they developed.

It is important to note that the three blockchain applications—data storage and collection, exchange, and validation—and the two resulting benefits for LCA, enhanced data reliability and traceability, are closely related and frequently mentioned together in the included studies due to their strong interdependence.

In the impact assessment phase, two key applications were identified: big data analytics and automated LCA calculation. Regarding big data analytics, Asif and Gill (2022) suggest that combining IoT and blockchain allows for the collection of large datasets, enabling LCA researchers and practitioners to utilize big data analytics. This not only improves LCA results but also provides opportunities for environmental improvements during the design phase of new products. Concerning automated LCA calculation, Lin et al. (2021) propose a blockchain-based LCA framework where each node automatically performs an LCA calculation whenever the required material flows are provided.

In the interpretation phase, blockchain can assist in identifying significant issues. Further, the application big data analytics, enabled by blockchain and suggested by Asif and Gill (2022), can also be viewed as part of the interpretation phase, as it can provide recommendations for environmental improvements during the design of new products. Carrières et al. (2022), in their study on blockchain's role in enhancing LCA in the textile industry, conclude that blockchain-enabled collection and storage of specific data improves the understanding and characterization of products. This, in turn, can enhance environmental sustainability and support sustainable design choices. Therefore, the specific data collected via blockchain also contributes to the interpretation phase by helping to identify so-called hot spots (major impact contributors). Based on the previous description, the following example illustrates how specific data collected via blockchain for LCA could support decision making in the context of sustainability. Given that the complete supply chain takes part in data sharing via blockchain for sustainability assessments, decision-makers responsible for sustainable sourcing of raw materials and intermediate products can analyze these data to assess environmental impacts. If the analysis identifies higher-than-average emissions or environmental impacts from a specific supplier, the company can decide to switch suppliers, thereby improving the overall sustainability and reducing the environmental footprint of the final product.

Table S2 in the supplementary information summarizes all the blockchain-focused papers from the literature sample.

4.3 Life cycle assessment and artificial intelligence

The application of AI in LCA represents the largest share of papers included in this study, with a total of 78 papers focusing on this area. AI is utilized across all four phases according to ISO 14040/44 standards. Figure 4 provides an overview of AI applications within the LCA framework as identified in the qualitative content analysis described in Sect. 3. Given that the application "Prediction of life cycle assessment results" is particularly prominent in the literature, with 35 papers addressing it, this category is discussed separately. Section 4.3.1, "Life cycle assessment and artificial intelligence in general," covers papers focusing on AI applications in LCA, excluding those specifically dealing with the prediction of LCA results. Section 4.3.2, "Prediction of life cycle assessment results with artificial intelligence," summarizes papers on LCA prediction and estimation. Tables S3 and S4 in the supplementary information provide summaries of all AI-focused papers in the sample.

4.3.1 Life cycle assessment and artificial intelligence in general

In the goal and scope definition phase, AI and machine learning can be utilized to determine assumptions. For example, Ji et al. (2021) and Østergaard et al. (2018) use machine learning to determine the temporal scope of buildings.

In the inventory analysis phase, AI offers various applications. Lopez-Andres et al. (2018) use AI for allocation in their LCA of chicken meat production, comparing three allocation methods: the mass method, artificial neural networks (ANNs), and stepwise regression, all of which produce similar results.

AI can also aid in estimating missing values for LCI data. For instance, Hou et al. (2018) estimate missing unit process data using similarity-based link prediction from known similar processes, showing that accurate estimates are possible when less than 5% of data is missing. Complementary to the similarity-based approach, Zhao et al. (2021) develop a method for estimating missing unit process data by applying a decision tree-based supervised learning approach for the case that more than 5% of data is missing. Khadem et al. (2022) focus on optimizing ANN design for predicting CO_2-eq emission streams, using heuristics and a genetic algorithm to create optimal neural networks, which they test by predicting data gaps in the Canadian fuel LCI database, GHGenius.

In the field of chemicals, Wernet et al. (2008) develop a model for estimating inventory data based on molecular structure, and Wernet et al. (2012) compare various LCI estimation methods for chemicals, including one based on an ANN.

Additional approaches to handling missing data in the inventory analysis phase include Meron et al. (2020), who propose a methodology for selecting a proxy dataset for a specific site from available datasets, and Dai et al. (2022), who develop a framework for obtaining the best-fit secondary data using a Gaussian process regression (GPR) model.

AI and machine learning can also be used in the inventory analysis phase to create data-driven models. Li et al. (2017) propose a modular LCA framework that integrates both knowledge-based and data-driven models. They argue that a machine learning-based, data-driven approach can enhance traditional knowledge-based modeling methods. They illustrate their approach with an example involving an injection molding process, using a Bayesian network. Froemelt et al. (2018) investigate the environmental impacts of various household archetypes in Switzerland. They use a combination of a self-organizing map, an unsupervised machine learning algorithm, and a clustering algorithm to derive these archetypes. In a follow-up study, Froemelt et al. (2020) develop a regionalized bottom-up model to assess the environmental profiles of individual households in specific regions, integrating three of their previously developed models.

In addition to estimating missing data, AI and machine learning are also used for generating LCI data within the LCA framework. Liao et al. (2020) focus on biorefineries, employing kinetic-based process simulation and an ANN to generate LCI data for activated carbon produced from different types of woody biomass. Cheng et al. (2020b) use machine learning models to predict product yields and characteristics from the hydrothermal treatment of various feedstocks. These predictions serve as inputs for their LCA model to calculate net global warming potential and energy return on investment. The topic of biorefineries is particularly prominent in this category, with several studies dedicated to creating LCI data for biorefinery processes (Olafasakin et al. 2021; Huntington et al. 2023; Hajabdollahi Ouderji et al. 2023; Cheng et al. 2020a; Cheng et al. 2021; Karka et al. 2022).

Nabavi-Pelesaraei et al. (2016) use an ANN to predict kiwi production yields and greenhouse gas emissions. Additional examples of LCI prediction in agriculture include potato, wheat, and lentil production (Khoshnevisan et al. 2014b; Khoshnevisan et al. 2013; Elhami et al. 2017).

In the automotive sector, AI applications for determining LCI include estimating truck fuel consumption, predicting fuel cell degradation, and creating a micro energy consumption and carbon emission model for taxis (Perrotta et al. 2018; Raeesi et al. 2021; Li et al. 2022).

In the built environment, AI is used to predict energy consumption and carbon emissions throughout the life cycle (Xikai et al. 2019; Wang and Shen 2013).

Two additional applications of AI in the inventory analysis phase are highlighted by Adedeji et al. (2020), who present a roadmap for utilizing AI in this phase, particularly focusing on renewable energy systems. They propose employing AI for LCI in the context of data mining and data structuring. Similarly, Shou and Domenech (2022) suggest integrating AI into their blockchain-based LCA framework as a potential approach for data processing in the inventory analysis phase. Finally, AI can be used to estimate hourly-day-ahead electricity consumption mixes (Cornago et al. 2020).

In the impact assessment phase, AI and machine learning offer several applications. For instance, machine learning can be used to estimate missing characterization factors. Slapnik et al. (2015) calculate regional normalization factors for Slovenia and use a linear regression machine learning model to estimate missing characterization factors. Hou et al. (2020) apply machine learning to estimate the hazardous concentration for 50% of chemicals, using physical–chemical properties and classification as model inputs, which allows for the determination of ecotoxicity characterization factors. Marvuglia et al. (2015a) explore the use of machine learning for toxicity characterization in LCA, focusing on the USEtox database. They aim to explore the input space, which includes substance-specific properties, and develop an automatic selection strategy to identify particularly informative variables. They use machine learning models, such as general regression neural networks, to estimate 13 USEtox output factors based on nine input variables. Marvuglia et al. (2015b) also focus on the USEtox database, performing variable selection to identify informative variables for toxicity characterization based on molecular properties. Song et al. (2022) develop species sensitivity distributions (SSD) for organic chemicals using machine learning techniques, which can be used in LCA to derive ecotoxicity characterization factors. Servien, Latrille et al. (2022a) propose a machine learning-based method for calculating characterization factors using molecular descriptors, and Servien, Leenknecht et al. (2022b) apply this model in the context of wastewater treatment. Additionally, Gust et al. (2015) concentrate on toxicity characterization in LCAs and explore the use of adverse outcome pathways to enable more accurate biological effects assessments within LCA. They investigate three approaches for developing quantitative adverse outcome pathways, including a probabilistic approach that also employs supervised machine learning models to create these pathways.

AI also offers potential applications in the interpretation phase of an LCA, particularly for identifying significant issues and quantifying uncertainty. Romeiko, Lee, et al. (2020b) use machine learning to pinpoint the most significant contributors among soil, weather, and farming practices regarding life cycle impacts. Prioux et al. (2023) support the analysis of pretreatment processes for lignocellulosic biomass in a circular economy context, employing unsupervised machine learning for dimension reduction, clustering, and environmental impact interpretation. More broadly, machine learning and AI can assist in identifying significant issues by determining the importance of predictor variables in the context of the application "Prediction of LCA results." Many machine learning models provide insights into which variables are most influential for specific LCA results. This information can help identify factors that could be adjusted to improve the environmental performance of a product or process. Examples of this application for identifying significant issues can be found in Karka et al. (2022), Romeiko et al. (2019), and Feng et al. (2019). More generally, Zhang et al. (2020) identify machine learning as a means to extract meaningful information from data recorded via their blockchain-based LCA framework.

In terms of uncertainty quantification, Dai et al. (2022) develop a framework to obtain the best-fit secondary data. They first determine correlations among production activities in existing data, considering temporal, geographic, and taxonomic dimensions. They then use covariance functions to quantify similarity and train a GPR model, which predicts secondary data. They argue that the GPR model's transparency and its prediction intervals provide an indication of uncertainty, which can be used for uncertainty analysis in LCA. They suggest that uncertainty shifts from input parameter uncertainty to GPR model uncertainty, altering the nature of uncertainty in secondary data. Ziyadi and Al-Qadi (2019) focus on model uncertainty in LCA. They explore three sources of model uncertainty—input uncertainty, parameter uncertainty, and model-form uncertainty—and apply machine learning as a surrogate to replace simulation software. This approach allows for the propagation of input variability through the system using interval analysis, independently of model-form uncertainty.

4.3.2 Prediction of life cycle assessment results with artificial intelligence

The most common application of AI in the analyzed publications is the prediction or estimation of LCA results. Early studies in this area focus on product development (Sousa et al. 2001; Park and Seo 2003; Seo et al. 2005; Sousa and Wallace 2006; Seo and Kim 2007). The earliest identified study, by Sousa et al. (2001), introduces an approximate model for streamlined LCA using high-level information typically available during the conceptual design phase. They train an ANN on product attributes and environmental impacts from previous LCA studies, enabling product design teams to estimate the environmental impacts of new product concepts based on these high-level attributes. More recent studies on predicting LCA results in product development include Wisthoff et al. (2016), who use a neural network to assess the environmental impacts of product attributes defined during the design phase, and Singh and Sarkar (2023), who develop an ANN approach to estimate carbon footprints based on life cycle attributes, such as product size, manufacturing process, and recyclability.

The use of AI and machine learning for predicting LCA results is particularly prominent in the context of agriculture, where various studies apply machine learning to predict the environmental impacts of agricultural production processes (e.g., wheat or tomato production) based on input variables such as electricity usage and chemical fertilizers (Khoshnevisan et al. 2014a; Elhami et al. 2017; Nabavi-Pelesaraei et al. 2018; Romeiko et al. 2019; Kaab et al. 2019; Romeiko et al. 2020a; Lee et al. 2020; Ghasemi-Mobtaker et al. 2022; Gundoshmian et al. 2022).

Machine learning for predicting LCA results is also frequently applied in the context of the built environment, especially during the early design phase of buildings, to help decision-makers achieve sustainability (Feng et al. 2019; Duprez et al. 2019; D'Amico et al. 2019; Płoszaj-Mazurek et al. 2020). Similarly, Sharif and Hammad (2019) develop a machine learning model to predict total energy consumption, life cycle costs, and LCA results for various renovation scenarios. Martínez-Rocamora et al. (2021) develop a methodology for generating environmental benchmarks for building typologies by combining AI with a building information modeling (BIM)-based LCA tool. They use the random forest algorithm to identify building features that significantly reduce environmental impact and to predict outcomes across various impact categories, including the total mass of materials. Related to the built environment, Onyelowe et al. (2022) focus on predicting the 28-day compressive strength and environmental impact of fly ash-based concrete, while Alam et al. (2021) create models to quantify CO_2 emissions from pavement construction in Canada. Lastly, Koyamparambath et al. (2022) apply AI to predict environmental impacts based on environmental product declaration information, using construction products as a case study.

In the field of chemicals, Wernet et al. (2008) use linear regression and neural networks to develop molecular structure-based models for estimating inventory data and predicting LCA impact factors. Similarly, Song et al. (2017) employed neural networks to predict the life cycle impacts of chemicals based on molecular structure information. Kleinekorte et al. (2019) develop a pathway-specific LCA prediction method using molecular descriptors as well as process descriptors. Zhu et al. (2020) propose a framework for making pharmaceutical manufacturing more sustainable, using sitagliptin production as an example and applying deep learning to estimate the life cycle impacts of chemicals within their framework. Sun et al. (2023) also focus on predicting LCA outcomes, with particular attention to feature selection to enhance the accuracy and interpretability of machine learning models.

Connected to the electronics sector, Chiang et al. (2011) develop a design-for-environment methodology aimed at improving consumer electronic product development. They created a neural network-based approximate LCA model to estimate the quantities of hazardous chemical substances and energy consumption. Li et al. (2008) develop an ANN prediction and assessment model to complement existing LCA tools, focusing on predicting Eco-indicator 99 values using computer products as a case study.

In the context of the automotive industry, Akhshik et al. (2022) apply AI-based predictions of LCA results to explore the environmental benefits of replacing glass fiber composites with natural fiber-based composites.

In the field of biorefineries, Mayol et al. (2020) develop an AI-driven model to predict the environmental impact of an algal biorefinery.

Lastly, Satinet and Fouss (2022) take a different approach by creating a model that allows quick and easy sustainability assessments of clothing products. Their machine learning-based model extrapolates the results of existing LCAs, enabling online retailers to evaluate the sustainability of clothing items in their assortment with limited data.

4.4 Life cycle assessment and big data

Out of the 103 papers included in the analysis, 19 focused on the use of big data in LCA. Big data can be applied in various phases of the ISO 14040/44 framework, including the goal and scope definition, inventory analysis, impact assessment, and interpretation phases. Figure 5 provides a summary of the applications of big data in LCA, as identified through the qualitative content analysis described in Sect. 3. A detailed overview of the big data-focused papers from the literature sample can be found in Table S5 of the supplementary information.

The use of big data for machine learning appears in three phases of LCA, including the goal and scope definition, inventory analysis, and impact assessment phases. In the goal and scope definition phase, big data serves as input for machine learning models, influencing the boundaries of the LCA. For example, Ji et al. (2021) use big data in their machine learning model to predict the lifespan of buildings, which is relevant to setting the goal and scope of the LCA. Since lifespan prediction falls within this phase, using big data as input for the model aligns with this stage of the process. In the inventory analysis phase, two studies (Perrotta et al. 2018; Li et al. 2022) focus on the application of big data for machine learning to generate LCIs in the automotive sector. Finally, in the impact assessment phase, Feng et al. (2019) propose a machine learning method for environmental assessment during the early design stages of buildings, using extensive design datasets. This highlights how big data supports machine learning in assessing environmental impacts based on large-scale data.

Big data can also be used, and is present in the inventory analysis phase during data collection. Three studies explore the use of big data in LCA with a focus on shared mobility, where transportation big data was integrated into the LCA process (Wang and Sun 2022; Sun and Ertz 2021, 2022).

Additionally, three other studies examine various pretreatment processes and the role of big data in LCA for data collection purposes (Prioux et al. 2023; Belaud et al. 2019, 2022). In particular, Prioux et al. (2023) develop an approach to support the analysis and comparison of pretreatment processes for lignocellulosic biomass and other types of biomass. They propose a five-step data architecture inspired by massive data systems. These steps include data collection and extraction, data enrichment and storage, data processing, raw data analysis, and raw data visualization. Belaud et al. (2022) propose a method for using data mining to obtain foreground data from scientific articles and apply it to complete the LCI. They demonstrate the applicability of their method through a case study focused on lignocellulosic waste pretreatment. Belaud et al. (2019) investigate the integration of Industry 4.0 and big data for improving sustainability management in supply chain design, with a focus on the valorization of agricultural waste. They design a five-step approach, which includes goal setting, data architecture, sustainability analysis, sustainability visualization, and decision-making. In the second step of this approach, they develop a big data architecture consisting of five sub-steps, which provides all the data necessary for the subsequent sustainability analysis.

The remaining studies on data collection are more heterogeneous. Yuan and Jin (2015) focus on the life cycle energy consumption assessment of buildings and the associated challenge posed by the large amount of data to be processed. They argue that with the emergence of big data technology and BIM, there is an opportunity to leverage these tools for a comprehensive life cycle energy assessment. To this end, they develop a building life cycle energy consumption framework based on summarizing the characteristics and demands of the necessary data. Beier et al. (2022) investigate potential big data use cases in environmental management, using the German automotive industry as a case study. Regarding LCA, they identify optimized LCA calculations as a potential use case for big data. Adedeji et al. (2020) highlight big data from continuous and real-time sensing as a challenge for data management. Conversely, Asif and Gill (2022) view the continuous sensing of data that results in big data as an opportunity for big data analytics. Similarly, Li et al. (2017) see big data as an opportunity to apply data-driven modeling techniques for LCA. Mashhadi and Behdad (2018) identify the extensive data collection enabled through smart manufacturing and the resulting new databases as a feature of future LCAs. Additionally, the big BOM concept developed by Tao et al. (2014) can be viewed as an approach to handling big data in the inventory analysis phase, as it aims to integrate data sensed via IoT into existing enterprise information systems.

In the impact assessment phase, big data can be used for enhanced analysis. Zhang et al. (2020) indicate that combining big data with supply chain analytics in this phase can lead to insights that facilitate decision-making, as the big data generated via IoT was previously unavailable for such insights.

Related to the interpretation phase, big data can help address uncertainty issues by reducing the uncertainty of LCA results. Ross and Cheah (2019) study the usefulness of high-granularity use-stage data for LCA through a case study focusing on AC systems. They use big data obtained from pervasive sensing and conclude that this high-granularity data reduces the overall uncertainty of LCA results and increases confidence in the results.

4.5 Integrated framework to combine digital technologies with life cycle assessment

The results so far show that digital technologies are rarely used in isolation but are often combined with other technologies. This section introduces a framework for the potential combination of the four digital technologies, where the development of the framework is described in detail in Sect. 3. The framework is illustrated in Fig. 6, depicting a simplified supply chain. The technologies and their previously identified potential applications are shown to the left of the supply chain. The producer is chosen as the focal company and is marked in red as the "*LCA conducting company*" at the center of the framework. The supply chain is included to illustrate the applications of data collection, storage, and exchange. The raw material extractor, the supplier, the consumer, and the end-of-life actor are assigned indices (i, j, k, h). These indices were introduced because a focal company will have multiple suppliers (j), who receive resources from various raw material extractors (i). The producer may also sell the product to numerous consumers (k), leading to products potentially reaching multiple end-of-life actors (h).

Data collection can primarily be carried out via IoT technologies, as proposed by Garcia-Muiña et al. (2018). However, not all LCA-relevant data can be collected through IoT technologies, as noted by Chit et al. (2021), who mention difficulties, for example, in the raw material extraction stage.

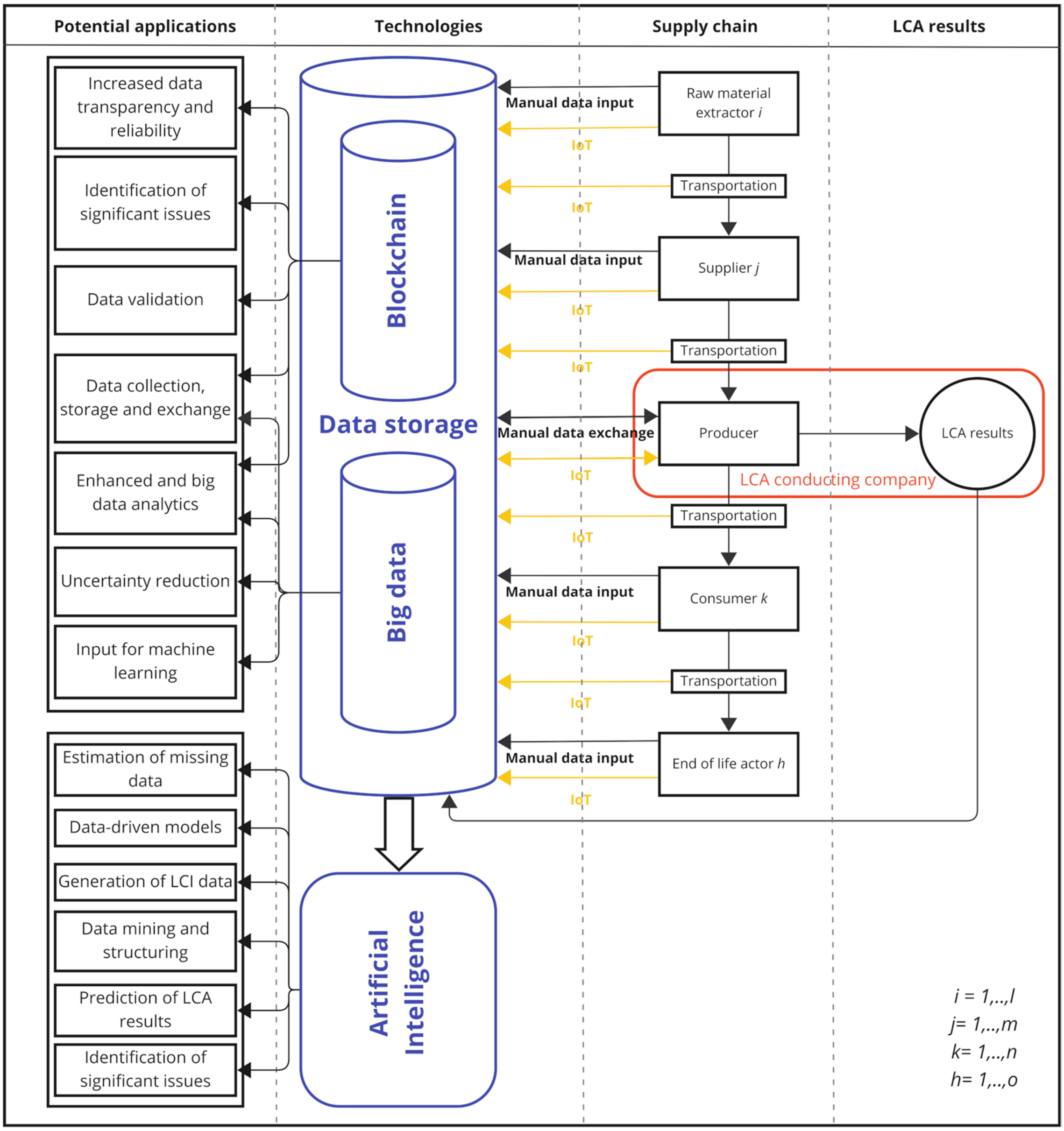

Fig. 6 Proposed combination framework

Therefore, the framework also includes the possibility of data being transferred manually to the corresponding data storage. Specifically, different supply chain actors (raw material extractor, supplier, producer, consumer, and end-of-life actor) enter LCA-relevant data either via IoT or manually into the corresponding database. The producer, as the LCA conducting company in this framework, plays a special role: it not only provides data for the LCA calculation itself but also extracts data provided by other actors for the LCA calculation. This is illustrated by the arrows pointing in both directions for the producer. For the transportation stage, it is assumed that no manual data entries are necessary, as relevant data could easily be obtained with the help of IoT devices (e.g., Shakhbulatov et al. (2019)).

As shown in Fig. 6, data storage can be supported by blockchain and big data technologies, both separately and in combination. The integration of IoT with blockchain is proposed by several authors, such as Asif and Gill (2022). Using blockchain as a storage solution for collected primary data enables additional applications. Blockchain-based data collection, storage, and exchange can enhance the reliability and traceability of LCA data, thus facilitating data validation in the inventory analysis phase. Moreover, blockchain data can assist in identifying significant issues during the interpretation phase of an LCA (Carrières et al. 2022). Additionally, the combination of IoT and blockchain can generate big data (e.g., Asif and Gill 2022; Zhang et al. 2020). Therefore, the creation of big data through the integration of IoT and blockchain opens up more potential applications. However, it must be noted that big data can also be created and stored without utilizing blockchain technology. The generation of big data enables the application of enhanced and big data analytics (e.g., Asif and Gill 2022; Zhang et al. 2020). Additionally, big data can be used to reduce uncertainty in LCA results, as suggested by Ross and Cheah (2019). Furthermore, the collected big data can serve as input for various machine learning and AI applications (e.g., Perrotta et al. 2018; Feng et al. 2019). In addition to these examples explicitly mentioning the use of big data for training machine learning models, other machine learning applications can also benefit from the generated data. For instance, the vast amount of available primary data facilitates the estimation of missing data in the inventory analysis phase. An example of this is the method proposed by Meron et al. (2020), which focuses on selecting the most appropriate site-specific proxy LCI data set for LCA calculations, rather than relying on generic country-specific data sets. In this case, greater availability of LCI data would provide a larger pool from which suitable site-specific data could be selected, leading to improved LCA results. Similarly, the estimation of missing data, as introduced by Dai et al. (2022), would benefit from the availability of more primary data being available and also the prediction of LCA results would be improved by the higher amount of data being available through the proposed approach. Additionally, the creation of data-driven models for LCA (e.g., Li et al. 2017) can be facilitated by data collected via IoT from various processes. Furthermore, AI techniques can be employed for data structuring and data mining of the collected big data (Adedeji et al. 2020). Finally, machine learning and AI techniques can be used to detect patterns and identify hot spots and other significant issues in both the collected data and the LCA results (e.g., Romeiko, Lee et al. 2020b). This identification of influencing factors can also be combined with the prediction of LCA outcomes, allowing for the assessment of which predictor variables have a greater influence on the results (e.g., Karka et al. 2022).

Overall, this framework illustrates how the combined use of the four investigated technologies can enable various applications. IoT is utilized for data collection and to transfer the data in the storage. The data storage can leverage blockchain technology, and the collected data can generate big data, facilitating numerous applications. Additionally, AI and machine learning techniques can be employed to analyze the collected data or use it to train algorithms for predicting LCI data or LCA results. This framework provides an overview of how a potential joint deployment of the four technologies could be structured within a simplified supply chain.

Companies that wish to implement digital technologies in their LCA processes should first assess their current processes and identify challenges (e.g., data collection, exchange, or handling missing data). Based on the identified challenges, in the second step, the company can explore existing digital solutions. If none are available, the company can start developing or commissioning the development of a custom digital solution. After a potential solution has been identified or created, the company may implement the digital technology solution in the next step. During the implementation stage, it is important to involve all relevant stakeholders and ensure that the knowledge and resources are available for a successful implementation. Reference is made to Sect. 5, which offers further insights into challenges and solutions during the implementation of digital technologies for LCA. After the successful implementation of the digital technology for LCA, a continuous review and improvement process is crucial to guarantee the correctness of the data collected or estimated and to ensure that technological improvements are timely detected and implemented.

5 Discussion

This paper presents a systematic literature review to identify the use of four selected digital technologies for LCA and proposes a framework for combining these technologies to improve the execution of LCAs. The applications of the four technologies—IoT, blockchain, AI, and big data—were assigned to the different LCA phases according to ISO 14040/44 standards. The strong focus on the application of digital technologies within the ISO 14040/44 framework is unique to this paper and differentiates it from previous work, such as the review by Ghoroghi et al. (2022), which focuses solely on AI and LCA. This study enhances the work of Ghoroghi et al. (2022) by not only concentrating on the contribution of AI technologies to the LCA framework according to ISO 14040/44, rather than the mainly technology- and algorithm-focused approach they adopted, but also by providing insights into potential combinations of AI with other digital technologies to further improve and facilitate the conduct of LCAs. Similarly, this study advances the work of Karaszewski et al. (2021) by identifying applications of blockchain for LCA in greater detail and by exploring possible combinations with other digital technologies.

5.1 Applications of digital technologies for LCA

Regarding RQ 1, which focuses on the identification of applications of the four digital technologies, three main findings emerged from the systematic review:

First, the literature review, detailed in Sect. 3, revealed that 17 studies focused on or mentioned IoT in the context of LCA, 14 studies mentioned blockchain, 78 studies discussed AI and machine learning, and 19 studies referred to the use of big data. There is a clear focus on the application of AI for LCA, likely due to the wide variety of possible AI applications compared to the more limited applications of the other three technologies. Furthermore, the analysis indicates that the most popular application across all investigated technologies is the prediction of LCA results, which is described separately in Sect. 4.3.2.

Second, regarding the implementation level, the study found that only four papers present applications that are already partly or broadly implemented, while 18 papers introduce conceptual applications and 81 papers include case illustrations. The relatively low number of papers describing broadly implemented applications might be due to several factors. Companies may be more advanced in implementing digital technologies to support sustainability management and LCAs but may withhold sharing their progress due to confidentiality concerns. Additionally, the adoption of digital technologies for LCA might still be in its early stages as companies may encounter difficulties with their implementation in the context of sustainability assessment. Although digital technologies might offer solutions to LCA-related challenges described in Sect. 2.1, companies might still face barriers related to digital transformation in general. According to Shahi and Sinha (2021), the main challenges companies face regarding digital transformation include a lack of vision (i.e., the absence of a clear strategy or objective), an organizational culture that is often resistant to change, inadequate skill sets, insufficient infrastructure, limited budgets, poor team collaboration, and data security concerns. Shahi and Sinha (2021) propose several potential solutions to these challenges: recruiting the right talent, adopting an agile methodology by breaking down larger goals into smaller tasks, embracing collaboration and information sharing across different functions within a company, and engaging all stakeholders. These general challenges and solutions can be adapted to the implementation of digital technologies for LCA. Firstly, to effectively implement digital technologies for sustainability management and LCA, a clear vision and strategy must be established. The overall goal should be broken down into smaller, manageable tasks to address any resistance to change within the organizational culture. Adequate funding is also crucial to ensure that the necessary talent is available and can be recruited if necessary and that suitable infrastructure is provided. Finally, involving all relevant stakeholders is essential to ensure a successful transformation process. In the case of LCA, involving stakeholders along the entire supply chain is crucial for obtaining the necessary data and ensuring joint data management and security. It is equally important to engage internal company stakeholders during the implementation of digital technologies for LCA. This involvement can help leverage potentially beneficial synergies, such as combining data collection for LCA via IoT with predictive maintenance. To address these challenges, the identified applications and the framework developed in this paper can provide valuable guidance. The results of this study can assist decision-makers in crafting effective strategies for integrating digital technologies into LCA processes and offer insights into how these technologies can be applied and combined advantageously. Furthermore, the study can help identify the required skill sets and provide orientation for employees involved in and affected by the change process, which may aid in overcoming barriers related to resistance to change.

Third, Fig. 7 summarizes the applications and key benefits derived from the four digital technologies investigated for LCA. The figure highlights the different applications, each of them mentioned together with the relevant phase in the ISO 14040/44 framework. In the case that applications are affecting the overall LCA and not only specific phases, no specific categorization is included. So far, IoT applications (described in detail in Sect. 4.1) have primarily been investigated for data collection purposes, where they enhance the availability of primary data, which, in turn, improves the reliability and accuracy of LCAs (e.g., Garcia-Muiña et al. 2018; Van Capelleveen et al. 2018). Blockchain applications (described in detail in Sect. 4.2) are more diverse. First, blockchain facilitates data storage, exchange, and validation (e.g., Asif and Gill 2022). Further, potential applications of blockchain include automated LCA calculations, big data analysis (e.g., Lin et al. 2021; Asif and Gill 2022) and support in identifying significant issues (e.g., Shou and Domenech 2022). Although in this study various advantages of using blockchain in the context of LCA are introduced, it also has to be mentioned that blockchain itself can have negative consequences for the environment. The negative environmental impact of blockchain technology can be significant, as highlighted by Stoll et al. (2019), who found that Bitcoin's annual electricity consumption reaches 45.8 TWh, contributing to carbon emissions between 22.0 and 22.9 MtCO_2 per year. Therefore, when considering blockchain applications, including in the context of LCA, it is crucial to take potential negative environmental consequences into account. AI applications (described in detail in Sect. 4.3) are the most diverse of the investigated technologies. AI can support in determining assumptions (e.g., Ji et al. 2021), estimating missing data (e.g., Hou et al. 2018), generating LCI data (e.g., Liao et al. 2020), mining and structuring of data (e.g., Adedeji et al. 2020), and in the creation of data-driven models (e.g., Li

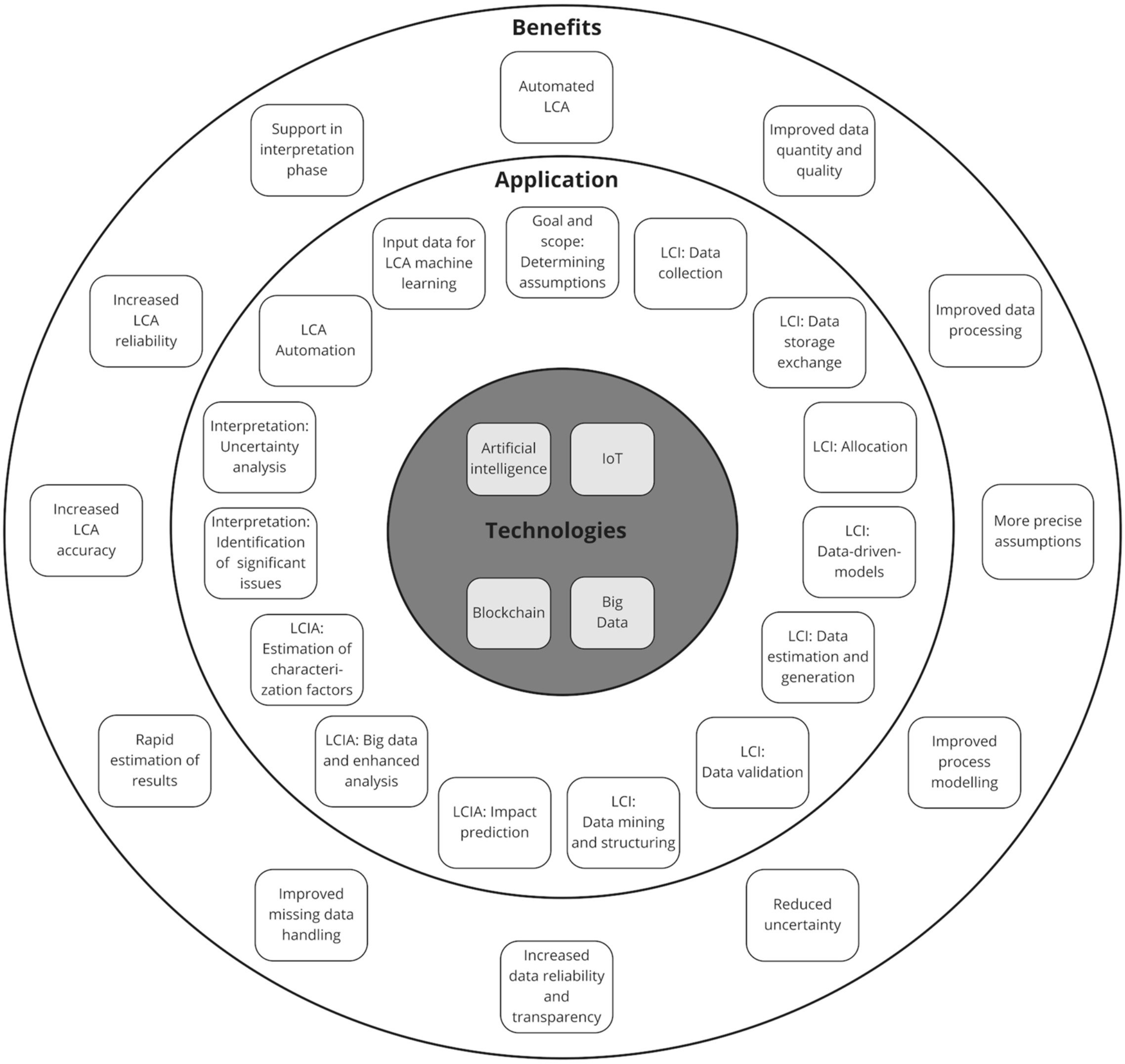

Fig. 7 Summary of applications of digital technologies for LCA and derived key benefits

et al. 2017). Additional applications of AI include the estimation of characterization factors (e.g., Slapnik et al. 2015) and overall LCA results (e.g., Akhshik et al. 2022), as well as support in identifying significant issues (e.g., Romeiko, Lee et al. 2020b) and uncertainty quantification (e.g., Dai et al. 2022). Lastly, big data applications (described in detail in Sect. 4.4) involve input for machine learning (e.g., Ji et al. 2021), enhancement of data collection (e.g., Belaud et al. 2022) enablement of improved analysis in the impact assessment phase (e.g., Zhang et al. 2020), and support in uncertainty analysis (Ross and Cheah 2019).

The key benefits derived from the identified applications of the four investigated digital technologies for LCA can be summarized as follows: Based on the integration of IoT and big data for data collection and blockchain for data storage and exchange both the quality and quantity of data available for LCA can be enhanced. AI and machine learning significantly enhance data processing capabilities, particularly in the inventory analysis phase. These technologies also enable higher precision in determining assumptions during the goal and scope phase. Process modeling can also be improved with the help of AI and supported by big data. Similarly, through AI and big data applications, the uncertainty of LCA results can be reduced. Applications of blockchain technology ensure greater data transparency and reliability, with the added potential that blockchain can enable the automation of LCA. AI applications are effective in addressing missing data in the inventory analysis and impact assessment phases. They also facilitate rapid estimations of LCA results, thereby making preliminary assessments more efficient. Big data can facilitate the different AI applications by providing the required input data to train the machine learning models. Overall, the application of the four investigated technologies—such as primary data collection via IoT, data storage and exchange through blockchain, and uncertainty reduction through big data—significantly enhances the accuracy and reliability of LCA results. Additionally, blockchain and AI further support the interpretation phase by identifying significant issues and hot spots of environmental impacts.

The summary of the applications reveals that the four investigated digital technologies each have distinct strengths and weaknesses when applied to LCA. IoT, for instance, offers significant potential for enhancing the availability of primary data but is limited to data collection purposes. Mansouri et al. (2023) highlight that data quality is essential for deriving business value from IoT implementations, a principle that also applies to its use in LCA. Consequently, a notable weakness of IoT is the quality of the collected data. Mansouri et al. (2023) identify potential data quality issues such as inconsistencies in the measured data, as well as duplicates and outliers detected by sensors. Blockchain offers various applications, with its primary strength lying in enhancing data reliability and traceability through secure data storage and exchange. However, implementing blockchain across the supply chain can be more complex than deploying IoT within a single company. This complexity arises from the involvement of multiple companies with varying interests, and challenges related to data security and confidentiality. Consequently, the increased implementation complexity and data privacy concerns can be considered relative weaknesses of blockchain. Additionally, the reliability of data stored on a blockchain depends on the quality of the input data. As Powell et al. (2022) note in the context of food supply chains, while blockchain ensures data immutability, the data can still be incorrect. Therefore, the dependency on input data quality for enabling effective blockchain applications represents another weakness of blockchain in the context of LCA. Therefore, the greatest strength of blockchain in LCA is its immutability, which supports various applications. However, this strength is fully realized only when additional measures are taken to ensure the quality of the stored data. AI's greatest strength lies in its ability to estimate and generate data that might otherwise be unavailable or of lower quality. However, a significant weakness is that AI and machine learning applications require existing data to train the models. The quality of the results and the feasibility of applying these techniques in LCA depend on the availability of quality data for training. Big data provides the essential data required for AI and machine learning applications, offering accurate and real-time information that can enhance LCA processes. The primary strength of big data is its capacity to supply vast amounts of information for various applications. However, this strength also presents a challenge, as managing the large volume, variety, and velocity of data requires advanced and tailored data management strategies. This can be a significant challenge for companies seeking to integrate big data into their LCA processes. In summary, while each technology has its own strengths and weaknesses, a combined approach might address some of these limitations. For example, data collected via IoT from a specific process, which generates big data, could be used to train AI and machine learning models to estimate missing data for similar processes.

5.2 Combined application of digital technologies for LCA

Regarding RQ 2, which examines the combined application of the four digital technologies for LCA, two main findings emerged.

Firstly, while several papers discuss the joint application of two or more of the investigated technologies, there is no existing literature proposing a combined application of all four technologies. Zhang et al. (2020) mention three relevant technologies but primarily focus on blockchain, with only a brief mention of AI and machine learning. Most studies that explore potential combinations tend to concentrate on pairs of technologies rather than integrating all four. Additionally, while papers on IoT, blockchain, or big data often discuss potential combinations with other technologies, those focused on AI applications rarely address the possibility of combining AI with other technologies. This is notable because data collected via IoT and big data could provide valuable input for machine learning applications. Although some studies take advantage of this potential, most do not. This may be due to the fact that many AI and machine learning applications are still in the early stages of implementation and are often presented as case studies. As a result, these applications may not yet be advanced enough to incorporate IoT and big data effectively. Nevertheless, there is a noticeable gap in research, particularly concerning the joint application of digital technologies for LCA. This gap is especially evident in studies that combine AI with other digital technologies within the context of LCA. Additionally, many of the identified studies do not delve deeply into how these technologies could be combined in practice or the challenges that might arise during the implementation process. Thus, there is also a lack of research focused on the joint implementation process of digital technologies.

Secondly, the identified combinations were used to construct a framework for integrating the four investigated technologies to enhance the execution of LCAs within an organization or company (Fig. 6). The framework is designed with a focus on the producer of a product or service and illustrates an exemplary supply chain. It emphasizes the collection and storage of data via IoT, blockchain, and big data applications, and how this data can be utilized for AI applications. This research aims to address the previously mentioned gap in studies regarding the joint application of digital technologies for LCA by proposing a framework that demonstrates a potential integration of these technologies. Therefore, the developed framework serves as a starting point for further research into the joint application of digital technologies in LCA, with a particular focus on best practices for implementation and the challenges that may arise. In this context, data privacy and the willingness to share data are also critical considerations. Although the proposed framework for integrating the four technologies could enhance LCA processes, it relies on the assumption that all involved actors are willing to share LCA-relevant data. This may not always be the case, as life cycle inventories often contain sensitive information about production processes and are frequently shared only in aggregated forms (Kuczenski et al. 2017). To fully realize the potential of digital technologies in facilitating LCAs, future research needs to address data privacy issues and the challenges associated with data sharing, as explored in studies such as those by Kuczenski et al. (2017) and Ghose (2024).

5.3 Limitations

This study has several limitations. Firstly, the scope of the review—limited to selected search terms and scientific literature—may be considered a constraint. While industry might have more advanced concepts for applying digital technologies to LCA, the assumption was made that scientific literature provides more established concepts for integrating these technologies compared to grey literature. Additionally, detailed background information is essential for understanding how digital technologies are applied in LCA, which might not always be available in company-specific applications. Therefore, focusing on scientific literature was deemed appropriate. Secondly, the development of the combination framework was based solely on literature, and empirical insights could have been valuable for assessing the feasibility of implementing the proposed framework. Despite this, the extensive number of papers reviewed provided a solid theoretical foundation, yielding valuable insights into the combined application of the four technologies. Finally, the study's focus on only the four investigated technologies can also be seen as a limitation. However, given the importance of these technologies within the European Union's regulatory context, this selection is considered to offer significant insights.

5.4 Further research

The literature review also suggests several directions for further research. Beyond the focus on companies and streamlined supply chains, it would be valuable to investigate other potential applications of the intersection between the four digital technologies and LCA. For instance, exploring perspectives from LCI data providers, LCA researchers, or the general public could provide additional insights. While this study concentrated on the company perspective, many of the reviewed studies offer only conceptual or illustrative examples. Further research could explore why broader implementation of LCA-related applications is not yet widespread. Specifically, investigating barriers to wider adoption and identifying supporting measures for the implementation of IoT, blockchain, big data, and AI in LCA could be beneficial. Additionally, further research could identify other potential applications of digital technologies within the LCA framework, such as exploring the extended use of AI technology in the interpretation phase. Finally, empirical studies could provide valuable insights and potential adaptations to the proposed framework.

6 Conclusions

Although LCA has seen continuous improvements over the years and many challenges have been addressed, some remain. The emergence of digital technologies and their growing implementation within companies offers new opportunities to overcome these LCA-related challenges. This paper examines the application of digital technologies in LCA through a systematic literature review and maps the identified applications to the LCA phases as defined by the ISO 14040/44 standards. The study proposes a framework for integrating the four investigated technologies based on the literature review. The findings indicate that AI and machine learning are the most frequently used technologies in the context of LCA, with 78 out of 103 papers discussing these technologies. IoT and big data follow, featured with 17 and respective 19 papers. Blockchain applications are discussed in 14 papers. AI and blockchain each have various applications across all four LCA phases. For AI, the most commonly mentioned application is the prediction of LCA results. For blockchain, the enhancement of data reliability and traceability is the most frequently cited application. Big data is also applicable in all four phases of the ISO 14040/44 framework, with its most common use being data collection. The IoT is exclusively utilized for data collection in the inventory analysis phase. Literature explores various potential combinations, such as using big data as input for machine learning, combining IoT and blockchain to ensure reliable and traceable primary data, and employing AI to analyze data collected via IoT or blockchain. The literature review leads to several implications. For research, the paper provides an overview of current studies on the application of digital technologies in LCA and can serve as a foundation for future investigations into the integration of these technologies into LCA. For practitioners and LCA users, the paper offers an overview of available LCA-related applications, assisting in the identification of tools suitable for addressing specific LCA challenges. The practical relevance of this research lies in providing LCA practitioners and sustainability managers with a basis for selecting appropriate digital technologies for conducting LCAs. Researchers should focus on overcoming challenges associated with implementing digital technologies for LCA and exploring how emerging technologies can be integrated. Practitioners should concentrate on incorporating available digital technologies into their LCA processes to enhance the reliability, validity, and real-world accuracy of LCA results. This can be achieved by utilizing more primary data (e.g., IoT or blockchain), making more realistic assumptions (e.g., AI), or reducing uncertainty (e.g., big data).

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11367-024-02409-4.

Acknowledgements The financial support provided by the Austrian Federal Ministry for Digital and Economic Affairs, the National Foundation for Research, Technology, and Development, the Christian Doppler Research Association, and the Austrian Research Promotion Agency (FFG) is gratefully acknowledged.

Funding Open access funding provided by University of Graz.

Data Availability Data will be made available upon request.

Declarations

Competing interest The authors have no competing interests to declare that are relevant to the content of this article.

References

Adedeji PA, Akinlabi SA, Madushele N, Olatunji OO (2020) Potential roles of artificial intelligence in the LCI of renewable energy systems. Adv in Manuf Eng. https://doi.org/10.1007/978-981-15-5753-8_26. Lecture Notes in Mechanical Engineering

Akhshik M, Bilton A, Tjong J, Singh CV, Faruk O, Sain M (2022) Prediction of greenhouse gas emissions reductions via machine learning algorithms: toward an artificial intelligence-based life cycle assessment for automotive lightweighting. Sustain Mater Technol 31:e00370. https://doi.org/10.1016/j.susmat.2021.e00370

Alam MR, Hossain K, Bazan C (2021) Life cycle analysis for asphalt pavement in Canadian context: modelling and application. Int J Pavement Eng 23(8):2606–2620. https://doi.org/10.1080/10298436.2020.1866759

An J, Zou Z, Chen G, Sun Y, Liu R, Zheng L (2021) An IoT-based life cycle assessment platform of wind turbines. Sensors 21(4):1233. https://doi.org/10.3390/s21041233

Asif MS, Gill H (2022) Blockchain technology and green supply chain management (GSCM)—improving environmental and energy performance in multi-echelon supply chains. IOP Conf Ser: Earth Environ Sci 952:012006. https://doi.org/10.1088/1755-1315/952/1/012006

Asopa P, Purohit P, Nadikattu RR, Whig P (2021) Reducing carbon footprint for sustainable development of smart cities using IoT. 2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV). pp 361–367. https://doi.org/10.1109/ICICV50876.2021.9388466

Bamber N, Turner I, Arulnathan V, Zarger Ershadi S, Smart A, Pelletier N (2020) Comparing sources and analysis of uncertainty in consequential and attributional life cycle assessment: review of current practice and recommendations. Int J Life Cycle Assess 25:168–180. https://doi.org/10.1007/s11367-019-01663-1

Beier G, Kiefer J, Knopf J (2022) Potentials of big data for corporate environmental management: a case study from the German automotive industry. J Ind Ecol 26(1):336–349. https://doi.org/10.1111/jiec.13062

Belaud JP, Prioux N, Vialle C, Sablayrolles C (2019) Big data for agri-food 4.0: application to sustainability management for by-products supply chain. Comput Ind 111:41–50. https://doi.org/10.1016/j.compind.2019.06.006

Belaud JP, Prioux N, Vialle C, Buche P, Destercke S, Barakat A, Sablayrolles C (2022) Intensive data and knowledge-driven approach for sustainability analysis: application to lignocellulosic waste valorization processes. Waste Biomass Valor 13:583–598. https://doi.org/10.1007/s12649-021-01509-8

Berger S, Denner MS, Röglinger M (2018) The nature of digital technologies—development of a multi-layer taxonomy. In: Proceedings of the 26th European Conference on Information Systems (ECIS) (S. 1–18)

Carrières V, Lemieux AA, Margni M, Pellerin R, Cariou S (2022) Measuring the value of blockchain traceability in supporting LCA for textile products. Sustainability 14(4):2109. https://doi.org/10.3390/su14042109

Chandrakumar C, McLaren SJ (2018) Towards a comprehensive absolute sustainability assessment method for effective Earth system governance: defining key environmental indicators using an enhanced-DPSIR framework. Ecolog Indic 90:577–583. https://doi.org/10.1016/j.ecolind.2018.03.063

Cheng F, Luo H, Colosi LM (2020a) Slow pyrolysis as a platform for negative emissions technology: an integration of machine learning models, life cycle assessment, and economic analysis. Energy Convers Manage 223:113258. https://doi.org/10.1016/j.enconman.2020.113258

Cheng F, Porter MD, Colosi LM (2020b) Is hydrothermal treatment coupled with carbon capture and storage an energy-producing negative emissions technology? Energy Convers Manage 203:112252. https://doi.org/10.1016/j.enconman.2019.112252

Cheng F, Small AA, Colosi LM (2021) The levelized cost of negative CO2 emissions from thermochemical conversion of biomass coupled with carbon capture and storage. Energy Convers Manage 237:114115. https://doi.org/10.1016/j.enconman.2021.114115

Chiang T-A, Che ZH, Wang T-T (2011) A design for environment methodology for evaluation and improvement of derivative consumer electronic product development. J Sys Sci Sys Eng 20(3):260–274. https://doi.org/10.1007/s11518-011-5168-8

Chit TW, Ning L, Paliath NA, Long YM, Akhtar H, Shanshan Y (2021) IoT-enabled and data-driven sustainability evaluation framework for textile supply chain. 2021 IEEE 16th Conference on Industrial Electronics and Applications (ICIEA). pp 297–304. https://doi.org/10.1109/ICIEA51954.2021.9516314

Ciroth A, Muller S, Weidema B, Lesage P (2016) Empirically based uncertainty factors for the pedigree matrix in Ecoinvent. Int J Life Cycle Assess 21(9):1338–1348. https://doi.org/10.1007/s11367-013-0670-5

Cornago S, Vitali A, Brondi C, Low JSC (2020) Electricity technological mix forecasting for life cycle assessment aware scheduling. Procedia CIRP 90:268–273. https://doi.org/10.1016/j.procir.2020.01.099

Crosby M, Nachiappan PP, Verma S, Kalyanaraman V (2016) Blockchain technology: Beyond bitcoin. Appl Innov Rev 2:6–19

Curran MA (2014) Strengths and limitations of life cycle assessment. In Klöpffer W (ed), Background and future prospects in life cycle assessment. LCA Compendium—the complete world of life cycle assessment, pp 189–206. https://doi.org/10.1007/978-94-017-8697-3

D'Amico A, Ciulla G, Traverso M, Lo Brano V, Palumbo E (2019) Artificial neural networks to assess energy and environmental performance of buildings: an Italian case study. J Clean Prod 239:117993. https://doi.org/10.1016/j.jclepro.2019.117993

Dai T, Jordaan SM, Wemhoff AP (2022) Gaussian process regression as a replicable, streamlined approach to inventory and uncertainty analysis in life cycle assessment. Environ Sci Technol 56(6):3821–3829. https://doi.org/10.1021/acs.est.1c04252

Dorsemaine B, Gaulier JP, Wary JP, Kheir N, Urien P (2015) Internet of Things: a definition and taxonomy. Proceedings – NGMAST 2015: The 9th international conference on next generation mobile applications, services and technologies September 72–77. https://doi.org/10.1109/NGMAST.2015.71

Duprez S, Fouquet M, Herreros Q, Jusselme T (2019) Improving life cycle-based exploration methods by coupling sensitivity analysis and metamodels. Sustain Cities Soc 44:70. https://doi.org/10.1016/j.scs.2018.09.032

Ecoinvent (2024) Ecoinvent database (3.10) [Software]. https://ecoinvent.org. Last accessed: 12.09.2024

Elhami B, Khanali M, Akram A (2017) Combined application of artificial neural networks and life cycle assessment in lentil farming in Iran. Inf Process Agric 4(1):18–32. https://doi.org/10.1016/j.inpa.2016.10.004

European Commission (2020) Communication from the commission to the European parliament, the council, the European economic and social committee and the committee of the regions: a new circular economy action plan. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM%3A2020%3A98%3AFIN. Accessed 8 Nov 2024

Fantke P, Chiu WA, Aylward L, Judson R, Huang L, Jang S, Gouin T, Rhomberg L, Aurisano N, McKone T, Jolliet O (2021) Exposure and toxicity characterization of chemical emissions and chemicals in products: Global recommendations and implementation in USEtox. Int J Life Cycle Assess 26:899–915. https://doi.org/10.1007/s11367-021-01889-y

Fasel D, Meier A (2016) Was versteht man unter big data und NoSQL? In: Fasel D, Meier A (eds) Big Data: Grundlagen, Systeme und Nutzungspotenziale. Springer Vieweg, Wiesbaden, pp 3–16. https://doi.org/10.1007/978-3-658-11589-0_1

Feng K, Lu W, Wang Y (2019) Assessing environmental performance in early building design stage: an integrated parametric design and machine learning method. Sustain Cities Soc 50:101596. https://doi.org/10.1016/j.scs.2019.101596

Froemelt A, Dürrenmatt DJ, Hellweg S (2018) Using data mining to assess environmental impacts of household consumption behaviors. Environ Sci Technol 52(15):8467–8478. https://doi.org/10.1021/acs.est.8b01452

Froemelt A, Buffat R, Hellweg S (2020) Machine learning based modeling of households: a regionalized bottom-up approach to investigate consumption-induced environmental impacts. J of Ind Ecol 24(3):639–652. https://doi.org/10.1111/jiec.12969

Garcia-Muiña F, González-Sánchez R, Ferrari A, Settembre-Blundo D (2018) The paradigms of industry 4.0 and circular economy as enabling drivers for the competitiveness of businesses and territories: The case of an Italian ceramic tiles manufacturing company. Soc Sci 7(12):255. https://doi.org/10.3390/socsci7120255

Ghasemi-Mobtaker H, Kaab A, Rafiee S, Nabavi-Pelesaraei A (2022) A comparative of modeling techniques and life cycle assessment for prediction of output energy, economic profit, and global warming potential for wheat farms. Energy Rep 8:4922–4934. https://doi.org/10.1016/j.egyr.2022.03.184

Ghoroghi A, Rezgui Y, Petri I, Beach T (2022) Advances in application of machine learning to life cycle assessment: a literature review. Int J Life Cycle Assess 27:433–456. https://doi.org/10.1007/s11367-022-02030-3

Ghose A (2024) Can LCA be FAIR? Assessing the status quo and opportunities for FAIR data sharing. Int J Life Cycle Assess 29:733–744. https://doi.org/10.1007/s11367-024-02280-3

Gimpel H, Röglinger M (2015) Digital transformation: changes and chances—insights based on an empirical study. https://fim-rc.de/wp-content/uploads/2020/02/Fraunhofer-Studie_Digitale-Transformation.pdf. Accessed 8 Nov 2024

Gossen M, Rohde F, Santarius T (2021) A marriage story of digitalisation and sustainability? Ökologisches Wirtschaften –. Fachzeitschrift 36(O1):4–8

Gundoshmian TM, Ardabili S, Csaba M, Mosavi A (2022) Modeling and optimization of the oyster mushroom growth using artificial neural network: economic and environmental impacts. Math Biosci Eng 19(10):9749–9768. https://doi.org/10.3934/mbe.2022453

Gust KA, Collier ZA, Mayo ML, Stanley JK, Gong P, Chappell MA (2015) Limitations of toxicity characterization in life cycle assessment: can adverse outcome pathways provide a new foundation? Integr Environ Assess Manag 12(3):580–590. https://doi.org/10.1002/ieam.1708

Hajabdollahi Ouderji Z, Gupta R, Mckeown A, Yu Z, Smith C, Sloan W, You S (2023) Integration of anaerobic digestion with heat pump: machine learning-based technical and environmental assessment. Bioresour Technol 369:128485. https://doi.org/10.1016/j.biortech.2022.128485

Hauschild MZ, Rosenberg RK, Olsen SI (2018) Life cycle assessment—theory and practice. Springer, Cham Switzerland. https://doi.org/10.1007/978-3-319-56475-3

Hou P, Cai J, Qu S, Xu M (2018) Estimating missing unit process data in life cycle assessment using a similarity-based approach. Environ Sci Technol 52(9):5259–5267. https://doi.org/10.1021/acs.est.7b05366

Hou P, Jolliet O, Zhu J, Xu M (2020) Estimate ecotoxicity characterization factors for chemicals in life cycle assessment using machine learning models. Environ Int 135:105393. https://doi.org/10.1016/j.envint.2019.105393

Huntington T, Baral NR, Yang M, Sundstrom E, Scown CD (2023) Machine learning for surrogate process models of bioproduction pathways. Bioresour Technol 370:128528. https://doi.org/10.1016/j.biortech.2022.128528

Ingrao C, Evola S, Cantore P, De Bernardi P, Del Borghi A, Vesce E, Beltramo R (2021) The contribution of sensor-based equipment to life cycle assessment through improvement of data collection in the industry. Environ Impact Assess Rev 88:106569. https://doi.org/10.1016/j.eiar.2021.106569

Islam S, Apu KU (2024) Decentralized vs. centralized database solutions in blockchain: advantages, challenges, and use cases. Glob Mainstream J Innov Eng Emerg Technol 3(4):58–68. https://doi.org/10.62304/jieet.v3i04.195

ISO (2006a) ISO 14040:2006: Environmental management life cycle assessment principles and framework. [Standard]. https://www.iso.org/standard/37456.html. Accessed 8 Nov 2024

ISO (2006b) ISO 14044:2006: Environmental management life cycle assessment requirements and guidelines. [Standard]. https://www.iso.org/standard/38498.html. Accessed 8 Nov 2024

Ji S, Lee B, Yi MY (2021) Building life-span prediction for life cycle assessment and life cycle cost using machine learning: A big data

approach. Build Environ 205:108267. https://doi.org/10.1016/j.buildenv.2021.108267
Jolliet O, Saadé-Sbeih M, Shaked S, Jolliet A, Crettaz P (2015) Environ life cycle assess, 1st edn. CRC Press, Florida. https://doi.org/10.1201/b19138
Kaab A, Sharifi M, Mobli H, Nabavi-Pelesaraei A, Chau K (2019) Combined life cycle assessment and artificial intelligence for prediction of output energy and environmental impacts of sugarcane production. Sci of the Total Environ 664:1005–1019. https://doi.org/10.1016/j.scitotenv.2019.02.004
Karaszewski R, Modrzynski P, Müldür GT, Wójcik J (2021) Blockchain technology in life cycle assessment – new research trends. Energies 14:8292. https://doi.org/10.3390/en14248292
Karka P, Papadokonstantakis S, Kokossis A (2022) Digitizing sustainable process development: from ex-post to ex-ante LCA using machine-learning to evaluate bio-based process technologies ahead of detailed design. Chem Eng Sci 250:117339. https://doi.org/10.1016/j.ces.2021.117339
Khadem SA, Bensebaa F, Pelletier N (2022) Optimized feed-forward neural networks to address CO2-equivalent emissions data gaps—application to emissions prediction for unit processes of fuel life cycle inventories for Canadian provinces. J Clean Prod 332:130053. https://doi.org/10.1016/j.jclepro.2021.130053
Khoshnevisan B, Rafiee S, Omid M, Yousefi M, Movahedi M (2013) Modeling of energy consumption and GHG (greenhouse gas) emissions in wheat production in Esfahan province of Iran using artificial neural networks. Energy 52:333–338. https://doi.org/10.1016/j.energy.2013.01.028
Khoshnevisan B, Rafiee S, Omid M, Mousazadeh H, Clark S (2014a) Environmental impact assessment of tomato and cucumber cultivation in greenhouses using life cycle assessment and adaptive neuro-fuzzy inference system. J Clean Prod 73:183–192. https://doi.org/10.1016/j.jclepro.2013.09.057
Khoshnevisan B, Rafiee S, Omid M, Mousazadeh H, Rajaeifar MA (2014b) Application of artificial neural networks for prediction of output energy and GHG emissions in potato production in Iran. Agric Syst 123:120–127. https://doi.org/10.1016/j.agsy.2013.10.003
Kiemel S, Rietdorf C, Schutzbach M, Miehe R (2022) How to simplify life cycle assessment for industrial applications - a comprehensive review. Sustain 14(15704):15704. https://doi.org/10.3390/su142315704
Kleinekorte J, Kröger L, Leonhard K, Bardow A (2019) A neural network-based framework to predict process-specific environmental impacts. Comput Aided Chem Eng 46:1447–1452. https://doi.org/10.1016/B978-0-12-818634-3.50242-3
Koyamparambath A, Adibi N, Szablewski C, Adibi SA, Sonnemann G (2022) Implementing artificial intelligence techniques to predict environmental impacts: case of construction products. Sustainability 14(6):3699. https://doi.org/10.3390/su14063699
Kuczenski B, Sahin C, El Abbadi A (2017) Privacy-preserving aggregation in life cycle assessment. Environ Syst Decis 37:13–21. https://doi.org/10.1007/s10669-016-9620-7
Larsen VG, Tollin N, Sattrup PA, Birkved M, Holmboe T (2022) What are the challenges in assessing circular economy for the built environment? A literature review on integrating LCA, LCC and S-LCA in life cycle sustainability assessment, LCSA. J Build Eng 50:104203. https://doi.org/10.1016/j.jobe.2022.104203
Lee EK, Zhang W-J, Zhang X, Adler PR, Lin S, Feingold BJ, Khwaja HA, Romeiko XX (2020) Projecting life-cycle environmental impacts of corn production in the U.S. Midwest under future climate scenarios using a machine learning approach. Sci Total Environ 714:136697. https://doi.org/10.1016/j.scitotenv.2020.136697
Li Q, Wang Y (2021) Blockchain's role in supporting circular supply chains in the built environment. 2021 IEEE International Conference on Blockchain (Blockchain), pp 578–583. https://doi.org/10.1109/Blockchain53845.2021.00087
Li J, Wu Z, Zhang HC (2008) Application of neural network on environmental impact assessment tools. In Sustain Manuf 1(1/2):100–121. https://doi.org/10.1504/IJSM.2008.019229
Li Y, Zhang H, Roy U, Lee YT (2017) A data-driven approach for improving sustainability assessment in advanced manufacturing. 2017 IEEE International Conference on Big Data (Big Data) 1736–1745. https://doi.org/10.1109/BigData.2017.8258116
Li X, Chen B, Liu Q, Tao J (2022) Dynamic capture and estimation of energy-emission benefits for electric taxicabs with spatiotemporal big data and deep-learning-based microscopic model. In Zhang X, Ren H, Lu Y, Wang C (eds) Advances in Transdisciplinary engineering: Volume 23: Proceedings of the 3rd international conference on green energy, environment and sustainable development (GEESD2022). IOS Press. https://doi.org/10.3233/ATDE220307
Liao M, Kelley S, Yao Y (2020) Generating energy and greenhouse gas inventory data of activated carbon production using machine learning and kinetic based process simulation. ACS Sustain Chem Eng 8:1252–1261. https://doi.org/10.1021/acssuschemeng.9b06522
Lin X, Li X, Kulkarni S, Zhao F (2021) The application of blockchain-based life cycle assessment on an industrial supply chain. Sustainability 13(23):13332. https://doi.org/10.3390/su132313332
Lopez-Andres JJ, Aguilar-Lasserre AA, Morales-Mendoza LF, Azzaro-Pantel C, Perez-Gallardo JR, Rico-Contreras JO (2018) Environmental impact assessment of chicken meat production via an integrated methodology based on LCA, simulation and genetic algorithms. J Clean Prod 174:477–491. https://doi.org/10.1016/j.jclepro.2017.10.307
Luo Z, Zhu J, Sun T, Liu Y, Ren S, Tong H, Yu L, Fei X, Yin K (2022) Application of the IoT in the food supply chain—from the perspective of carbon mitigation. Environ Sci Techno 56(15):10567–10576. https://doi.org/10.1021/acs.est.2c02117
Ma J, Kim HM (2015) Predictive usage mining for life cycle assessment. Transp Res Part D: Transp Environ 38:125–143. https://doi.org/10.1016/j.trd.2015.04.022
Mansouri T, Sadeghi Moghadam MR, Monshizadeh F, Zareravasan A (2023) IoT data quality issues and potential solutions: a literature review. Comput J 66(3):615–625. https://doi.org/10.1093/comjnl/bxab183
Martínez-Rocamora A, Rivera-Gomez C, Galan-Marin C, Marrero M (2021) Environmental benchmarking of building typologies through BIM-based combinatorial case studies. Autom Constr 132:103980. https://doi.org/10.1016/j.autcon.2021.103980
Marvuglia A, Kanevski M, Benetto E (2015a) Machine learning for toxicity characterization of organic chemical emissions using USEtox database: Learning the structure of the input space. Environ Int 83:72–85. https://doi.org/10.1016/j.envint.2015.05.011
Marvuglia A, Leuenberger M, Kanevski M, Benetto E (2015b) Random forest for toxicity of chemical emissions: Features selection and uncertainty quantification. J Environ Account Manag 3(3):229–241. https://doi.org/10.5890/JEAM.2015.09.003
Mashhadi AR, Behdad S (2018) Ubiquitous life cycle assessment (U-LCA): a proposed concept for environmental and social impact assessment of industry 4.0. Manuf Lett 15:93–96. https://doi.org/10.1016/j.mfglet.2017.12.012
Mayol AP, San Juan JLG, Sybingco E, Bandala A, Dadios E, Ubando AT, Culaba AB, Chen WH, Chang JS (2020) Environmental impact prediction of microalgae to biofuels chains using artificial intelligence: a life cycle perspective. IOP Conf Ser: Earth Environ Sci 463(1):012011. https://doi.org/10.1088/1755-1315/463/1/012011
Mayring P (2014) Qualitative content analysis: theoretical foundation, basic procedures and software solution. Klagenfurt. https://nbn-resolving.org/urn:nbn:de:0168-ssoar-395173. Accessed 8 Nov 2024
Meron N, Blass V, Thoma G (2020) Selection of the most appropriate life cycle inventory dataset: new selection proxy methodology

and case study application. Int J Life Cycle Assess 25(4):771–783. https://doi.org/10.1007/s11367-019-01721-8

Miah JH, Griffiths A, McNeill R, Halvorson S, Schenker U, Espinoza-Orias N, Morse S, Yang A, Sadhukhan J (2018) A framework for increasing the availability of life cycle inventory data based on the role of multinational companies. Int J Life Cycle Assess 23(9):1744–1760. https://doi.org/10.1007/s11367-017-1391-y

Mishra S, Singh SP (2019) Carbon management framework for sustainable manufacturing using life cycle assessment, IoT and carbon sequestration. BIJ 28(5):1396–1409. https://doi.org/10.1108/BIJ-01-2019-0044

Moher D, Liberati A, Tetzlaff J, Altman DG (2009) Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. BMJ 339:b2535. https://doi.org/10.1136/bmj.b2535

Moutik B, Summerscales J, Graham-Jones J, Pemberton R (2023) Life cycle assessment research trends and implications: A bibliometric analysis. Sustainability 15(18):13408. https://doi.org/10.3390/su151813408

Mutel CL (2023) Cycling from data island to island. [Conference presentation] Brightcon 2023, Luxembourg, 18.09.2023

Nabavi-Pelesaraei A, Rafiee S, Hosseinzadeh-Bandbafha H, Shamshirband S (2016) Modeling energy consumption and greenhouse gas emissions for kiwifruit production using artificial neural networks. J Clean Prod 133:924–931. https://doi.org/10.1016/j.jclepro.2016.05.188

Nabavi-Pelesaraei A, Rafiee S, Mohtasebi SS, Hosseinzadeh-Bandbafha H, Chau K (2018) Integration of artificial intelligence methods and life cycle assessment to predict energy output and environmental impacts of paddy production. Sci Total Environ 631–632:1279–1294. https://doi.org/10.1016/j.scitotenv.2018.03.088

Nilsson NJ (2010) The quest for artificial intelligence: a history of ideas and achievements. Cambridge University Press, Cambridge. https://doi.org/10.1017/CBO9780511819346

Olafasakin O, Chang Y, Passalacqua A, Subramaniam S, Brown RC, Mba Wright M (2021) Machine learning reduced order model for cost and emission assessment of a pyrolysis system. Energy Fuels 35(12):9950–9960. https://doi.org/10.1021/acs.energyfuels.1c00490

Onyelowe KC, Kontoni D-PN, Ebid AM, Dabbaghi F, Soleymani A, Jahangir H, Nehdi ML (2022) Multi-objective optimization of sustainable concrete containing fly ash based on environmental and mechanical considerations. Buildings 12(7):948. https://doi.org/10.3390/buildings12070948

Østergaard N, Thorsted L, Miraglia S, Birkved M, Rasmussen FN, Birgisdóttir H, Kalbar P, Georgiadis S (2018) Data driven quantification of the temporal scope of building LCAs. Procedia CIRP 69:224–229. https://doi.org/10.1016/j.procir.2017.11.057

Park JH, Seo K-K (2003) Approximate life cycle assessment of product concepts using multiple regression analysis and artificial neural networks. KSME Int J 17(12):1969–1976. https://doi.org/10.1007/BF02982436

Perrotta F, Parry T, Neves LC, Mesgarpour M (2018) A machine learning approach for the estimation of fuel consumption related to road pavement rolling resistance for large fleets of trucks. The Sixth International Symposium on Life-Cycle Civil Engineering (IALCCE 2018). pp 2011–2015. https://nottingham-repository.worktribe.com/output/950687. Accessed 8 Nov 2024

Płoszaj-Mazurek M, Rynska E, Grochulska-Salak M (2020) Methods to optimize carbon footprint of buildings in regenerative architectural design with the use of machine learning, convolutional neural network, and parametric design. Energies 13(20):5289. https://doi.org/10.3390/en13205289

Powell W, Foth M, Cao S, Natanelov V (2022) Garbage in garbage out: the precarious link between IoT and blockchain in food supply chains. J Indust Info Integr 25:100261. https://doi.org/10.1016/j.jii.2021.100261

Prioux N, Ouaret R, Hetreux G, Belaud J-P (2023) Environmental assessment coupled with machine learning for circular economy. Clean Techn Environ Policy 25:689–702. https://doi.org/10.1007/s10098-022-02275-4

Raeesi M, Changizian S, Ahmadi P, Khoshnevisan A (2021) Performance analysis of a degraded PEM fuel cell stack for hydrogen passenger vehicles based on machine learning algorithms in real driving conditions. Energy Convers Manage 248:114793. https://doi.org/10.1016/j.enconman.2021.114793

Rolinck M, Gellrich S, Bode C, Mennenga M, Cerdas F, Friedrichs J, Herrmann C (2021) A concept for blockchain-based LCA and its application in the context of aircraft MRO. Procedia CIRP 98:394–399. https://doi.org/10.1016/j.procir.2021.01.123

Romeiko XX, Guo Z, Pang Y (2019) Comparison of support vector machine and gradient boosting regression tree for predicting spatially explicit life cycle global warming and eutrophication impacts: a case study in corn production. 2019 IEEE International Conference on Big Data (Big Data) 3277–3284. https://doi.org/10.1109/BigData47090.2019.9005581

Romeiko XX, Guo Z, Pang Y, Lee EK, Zhang X (2020a) Comparing machine learning approaches for predicting spatially explicit life cycle global warming and eutrophication impacts from corn production. Sustainability 12(4):1481. https://doi.org/10.3390/su12041481

Romeiko XX, Lee EK, Sorunmu Y, Zhang X (2020b) Spatially and temporally explicit life cycle environmental impacts of soybean production in the U.S. midwest. Environ Sci Technol 54(8):4758–4768. https://doi.org/10.1021/acs.est.9b06874

Ross SA, Cheah L (2019) Uncertainty quantification in life cycle assessments: exploring distribution choice and greater data granularity to characterize product use. J Ind Ecol 23(2):335–346. https://doi.org/10.1111/jiec.12742

Saavedra-Rubio K, Thonemann N, Crenna E, Lemoine B, Caliandro P, Laurent A (2022) Stepwise guidance for data collection in the life cycle inventory (LCI) phase: building technology-related LCI blocks. J Clean Prod 366:132903. https://doi.org/10.1016/j.jclepro.2022.132903

Satinet C, Fouss F (2022) A supervised machine learning classification framework for clothing products' sustainability. Sustainability 14(3):1334. https://doi.org/10.3390/su14031334

Seo K-K, Kim W-K (2007) Approximate life cycle assessment of product concepts using a hybrid genetic algorithm and neural network approach. In Szczuka MS, Howard D, Ślęzak D, Kim H, Kim T, Ko I, Lee G, Sloot PMA (eds) Advances in hybrid information technology (ICHIT 2006) lecture notes in computer science Vol 4413. Springer, Berlin Heidelberg, pp 258–268. https://doi.org/10.1007/978-3-540-77368-9_26

Seo K-K, Min S-H, Yoo H-W (2005) Artificial neural network based life cycle assessment model for product concepts using product classification method. In: Gervasi O, Gavrilova ML, Kumar V, Laganá A, Lee HP, Mun Y, Taniar D, Tan CJK (eds) Computational science and its applications—ICCSA 2005. ICCSA 2005. Lecture Notes in Computer Science, Vol 3483, pp 458–466. Springer, Berlin Heidelberg. https://doi.org/10.1007/11424925_49

Servien R, Latrille E, Patureau D, Hélias A (2022a) Machine learning models based on molecular descriptors to predict human and environmental toxicological factors in continental freshwater. Peer Comm J 2:e15. https://doi.org/10.24072/pcjournal.90

Servien R, Leenknecht C, Bonnot K, Rossard V, Latrille E, Mamy L, Benoit P, Hélias A, Patureau D (2022b) Improved impact assessment of micropollutants release from WWTPs. Case Stud Chem Environ Eng 5:100172. https://doi.org/10.1016/j.cscee.2021.100172

Shahi C, Sinha M (2021) Digital transformation: challenges faced by organizations and their potential solutions. Int J of Innov Science 13(1):17–33. https://doi.org/10.1108/IJIS-09-2020-0157

Shakhbulatov D, Arora A, Dong Z, Rojas-Cessa R (2019) Blockchain implementation for analysis of carbon footprint across food supply chain. 2019 IEEE International Conference on Blockchain (Blockchain), pp 546–551. https://doi.org/10.1109/Blockchain.2019.00079

Sharif SA, Hammad A (2019) Developing surrogate ANN for selecting near-optimal building energy renovation methods considering energy consumption, LCC and LCA. J Build Eng 25:100790. https://doi.org/10.1016/J.JOBE.2019.100790

Shojaei A, Wang J, Fenner A (2020) Exploring the feasibility of blockchain technology as an infrastructure for improving built asset sustainability. Built Environment Project and Asset Management 10(2):184–199. https://doi.org/10.1108/BEPAM-11-2018-0142

Shou M, Domenech T (2022) Integrating LCA and blockchain technology to promote circular fashion—a case study of leather handbags. J of Clean Prod 373:133557. https://doi.org/10.1016/j.jclepro.2022.133557

Singh PK, Sarkar P (2023) An artificial neural network tool to support the decision making of designers for environmentally conscious product development. Expert Syst Appl 212:118679. https://doi.org/10.1016/j.eswa.2022.118679

Slapnik M, Istenič D, Pintar M, Udovč A (2015) Extending life cycle assessment normalization factors and use of machine learning—a Slovenian case study. Ecolog Indic 50:161–172. https://doi.org/10.1016/j.ecolind.2014.10.028

Song R, Keller AA, Suh S (2017) Rapid life-cycle impact screening using artificial neural networks. Environ Sci & Technol 51(18):10777–10785. https://doi.org/10.1021/acs.est.7b02862

Song R, Li D, Chang A, Tao M, Qin Y, Keller AA, Suh S (2022) Accelerating the pace of ecotoxicological assessment using artificial intelligence. Ambio 51:598–610. https://doi.org/10.1007/s13280-021-01598-8

Sousa I, Wallace D (2006) Product classification to support approximate life-cycle assessment of design concepts. Technol Forecast Soc Chang 73(3):228–249. https://doi.org/10.1016/j.techfore.2004.03.007

Sousa I, Wallace D, Eisenhard JL (2001) Approximate life-cycle assessment of product concepts using learning systems. J Ind Ecolog 4(4):61–81. https://doi.org/10.1162/10881980052541954

Sphera (2024) GaBi databases [Software]. https://lcadatabase.sphera.com. Last accessed: 12.09.2024

Stoll C, Klaaßen L, Gallersdörfer U (2019) The Carbon Footprint of Bitcoin. Joule 3(7):1647–1661. https://doi.org/10.1016/j.joule.2019.05.012

Sun S, Ertz M (2021) Environmental impact of mutualized mobility: evidence from a life cycle perspective. Sci Total Env 772:145014. https://doi.org/10.1016/j.scitotenv.2021.145014

Sun S, Ertz M (2022) Can shared micromobility programs reduce greenhouse gas emissions: Evidence from urban transportation big data. Sustain Cities Soc 85:104045. https://doi.org/10.1016/j.scs.2022.104045

Sun Y, Wang X, Ren N, Liu Y, You S (2023) Improved machine learning models by data processing for predicting life-cycle environmental impacts of chemicals. Environ Sci Technol 57(8):3434–3444. https://doi.org/10.1021/acs.est.2c04945

Tao F, Zuo Y, Xu LD, Lv L, Zhang L (2014) Internet of Things and BOM-based life cycle assessment of energy-saving and emission-reduction of products. IEEE Trans Industr Inf 10(2):1252–1261. https://doi.org/10.1109/TII.2014.2306771

Tao X, Mao C, Xie F, Liu G, Xu P (2018) Greenhouse gas emission monitoring system for manufacturing prefabricated components. Autom Constr 93:361–374. https://doi.org/10.1016/j.autcon.2018.05.015

Teh D, Khan T, Corbitt B, Ong CE (2020) Sustainability strategy and blockchain-enabled life cycle assessment: A focus on materials industry. Environ Syst Decis 40(4):605–622. https://doi.org/10.1007/s10669-020-09761-4

Testa F, Nucci B, Tessitore S, Iraldo F, Daddi T (2016) Perceptions on LCA implementation: evidence from a survey on adopters and nonadopters in Italy. Int J Life Cycle Assess 21(10):1501–1513. https://doi.org/10.1007/s11367-016-1106-9

Van Capelleveen G, Pohl J, Fritsch A, Schien D (2018) The Footprint of Things: a hybrid approach towards the collection, storage and distribution of life cycle inventory data. 5th Int Conf Infor Commu Technol Sustain 52:350–364. https://doi.org/10.29007/8pnj

Wang E, Shen Z (2013) Lifecycle energy consumption prediction of residential buildings by incorporating longitudinal uncertainties. J Civil Eng Manag 19(1):161–171. https://doi.org/10.3846/13923730.2013.802744

Wang Y, Sun S (2022) Does large scale free-floating bike sharing really improve the sustainability of urban transportation? Empirical evidence from Beijing. Sustain Cities Soc 76:103533. https://doi.org/10.1016/j.scs.2021.103533

Wernet G, Hellweg S, Fischer U, Papadokonstantakis S, Hungerbühler K (2008) Molecular-structure-based models of chemical inventories using neural networks. Environ Sci Technol 42(17):6717–6722. https://doi.org/10.1021/es7022362

Wernet G, Hellweg S, Hungerbühler K (2012) A tiered approach to estimate inventory data and impacts of chemical products and mixtures. Int J Life Cycle Assess 17(6):720–728. https://doi.org/10.1007/s11367-012-0404-0

Wisthoff A, Ferrero V, Huynh T, DuPont B (2016) Quantifying the impact of sustainable product design decisions in the early design phase through machine learning. Proceedings of the ASME 2016 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference Vol 4: 21st Design for Manufacturing and the Life Cycle Conference; 10th International Conference on Micro- and Nanosystems. Charlotte, North Carolina, USA August 21–24 2016 V004T05A043. https://doi.org/10.1115/DETC2016-59586

Xikai M, Lixiong W, Jiwei L, Xiaoli Q, Tongyao W (2019) Comparison of regression models for estimation of carbon emissions during building's lifecycle using designing factors: a case study of residential buildings in Tianjin, China. Energy Build 204:109519. https://doi.org/10.1016/j.enbuild.2019.109519

Yuan Y, Jin Z (2015) Life cycle assessment of building energy in big-data era: theory and framework. 2015 International Conference on Network and Information Systems for Computers, pp 601–605. https://doi.org/10.1109/ICNISC.2015.130

Zargar S, Yao Y, Tu Q (2022) A review of inventory modeling methods for missing data in life cycle assessment. J of Ind Ecolog 26:1676–1689. https://doi.org/10.1111/jiec.13305

Zhang A, Zhong RY, Farooque M, Kang K, Venkatesh VG (2020) Blockchain-based life cycle assessment: An implementation framework and system architecture. Resour Conserv Recycl 152:104512. https://doi.org/10.1016/j.resconrec.2019.104512

Zhao B, Shuai C, Hou P, Qu S, Xu M (2021) Estimation of unit process data for life cycle assessment using a decision tree-based approach. Environ Sci Technol 55:8439–8446. https://doi.org/10.1021/acs.est.0c07484

Zhu X, Ho C-H, Wang X (2020) Application of life cycle assessment and machine learning for high-throughput screening of green chemical substitutes. ACS Sustain Chem Eng 8(30):11141–11151. https://doi.org/10.1021/acssuschemeng.0c02211

Ziyadi M, Al-Qadi IL (2019) Model uncertainty analysis using data analytics for life-cycle assessment (LCA) applications. Int J of Life Cycle Assess 24:945–959. https://doi.org/10.1007/s11367-018-1528-7